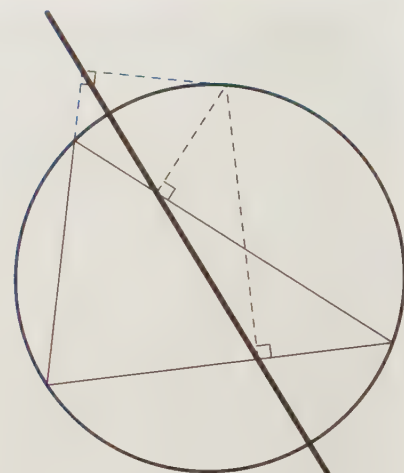




The Simson Line

At a round table there is no dispute about place. – Italian Proverb

CHAPTER 12

Circles and Angles

In this chapter we will explore the many ways in which angles and arc measures are related.

12.1 Inscribed Angles

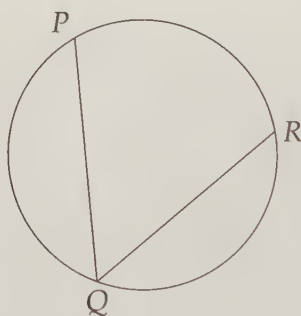
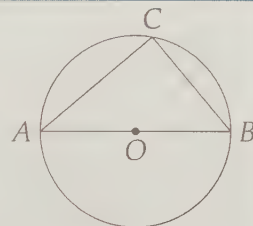


Figure 12.1: An Inscribed Angle

We say that $\angle PQR$ is **inscribed** in $\widehat{PR}$ in the diagram above because its vertex is on the circumference of the circle and its sides hit the circle at P and R . **Inscribed angles** are enormously useful tools in geometry; in this section we explore how inscribed angles are related to the arcs they cut off.

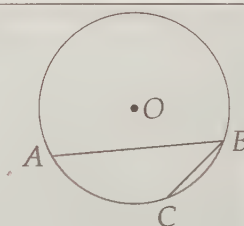
Problems

Problem 12.1: On page 175, we learned that the circumcenter of a right triangle is the midpoint of its hypotenuse. Thus, when we draw the circumcircle of a right triangle, the right angle is inscribed in a semicircle. In this problem, we investigate if it is true that any angle inscribed in a semicircle, such as $\angle ACB$ in the diagram at right, must be a right angle. Let O be the center of the circle.



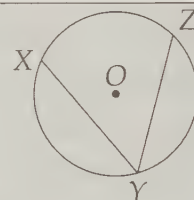
- Draw $\overline{OC}$. What do we know about $\triangle AOC$ and $\triangle BOC$?
- Let $\angle A = x$ and $\angle B = y$. What other angle measures x degrees? What other angle measures y degrees?
- Consider the sum of the angles in $\triangle ABC$ to show that $x + y = 90^\circ$.
- Must $\angle ACB$ be a right angle?

Problem 12.2: In this problem we will find inscribed $\angle ABC$ shown at right, given that $\widehat{AC} = 80^\circ$ and $\widehat{ACB} = 130^\circ$.



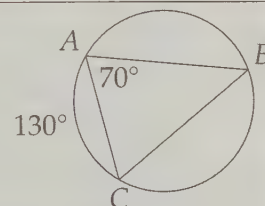
- Draw $\overline{OA}$, $\overline{OB}$, and $\overline{OC}$. Find $\angle AOC$ and $\angle COB$.
- Find $\angle OBC$ and $\angle OBA$.
- Use your answer to the previous part to find $\angle ABC$.
- Redo the problem with $\widehat{BC} = 64^\circ$.
- Make a guess about how we can figure out $\angle B$ from the arcs without going through all the steps above. Can you prove your guess?

Problem 12.3: In Problem 12.1, we showed that if an angle is inscribed in a 180° arc (a semicircle), the measure of the angle is $180^\circ/2 = 90^\circ$. In this problem we will prove that when $\angle XYZ$ is inscribed in $\widehat{XZ}$ as shown in the diagram (with O , the center of the circle, inside $\angle XYZ$), then $\angle Y = \widehat{XZ}/2$.

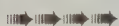


- Draw $\overline{OX}$, $\overline{OY}$, and $\overline{OZ}$. Let $\angle OXY = x$ and $\angle OZY = z$. Find $\angle XYZ$ in terms of x and z .
- Find $\angle XOY$, $\angle YOZ$, and $\angle XOZ$ in terms of x and z .
- Find $\widehat{XZ}$ in terms of x and z , then show that $\angle XYZ = \widehat{XZ}/2$.

Problem 12.4: Given that $\triangle ABC$ is inscribed in the circle as shown, $\angle A = 70^\circ$, and $\widehat{AC} = 130^\circ$, find $\angle C$.

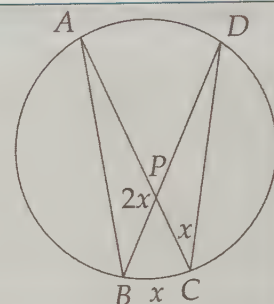


Extra! The primary question is not what do we know, but how do we know it.

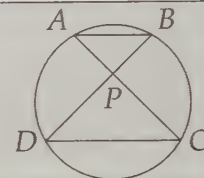


—Aristotle

Problem 12.5: Find x given that $\angle APB = 2x$, $\angle ACD = x$, and $\widehat{BC} = x$.



Problem 12.6: Chords $\overline{AC}$ and $\overline{BD}$ of a circle meet at P as shown. Given $AP = PB$, show that $\overline{AB} \parallel \overline{CD}$.



On page 175, we learned that when we draw the circumcircle of a right triangle, the right angle of the triangle cuts off a semicircle. We start this section by exploring whether the converse is true.

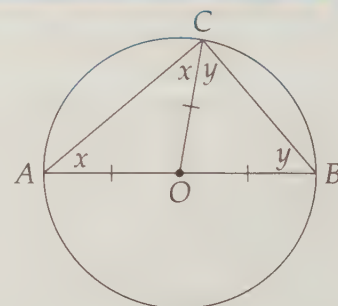
Problem 12.1: When an inscribed angle cuts off a 180° arc, we say that it is inscribed in a semicircle. Is such an angle necessarily a right angle?

Solution for Problem 12.1: Let $\angle ACB$ be inscribed in a semicircle. Since $\widehat{AB} = 180^\circ$, $\overline{AB}$ is a diameter of the circle. Therefore, the midpoint of $\overline{AB}$, point O , is the center of the circle. Drawing $\overline{OC}$ forms two isosceles triangles since $AO = BO = CO$.

We suspect $\angle ACB$ is 90° , so we let $\angle ACO = x$ and $\angle BCO = y$ and try to show that $x + y = 90^\circ$. Since $\triangle AOC$ and $\triangle BOC$ are isosceles, we have $\angle OAC = \angle OCA = x$ and $\angle OBC = \angle OCB = y$. From $\triangle ABC$, we have $\angle A + \angle ACB + \angle B = 180^\circ$, so

$$x + (x + y) + y = 180^\circ.$$

Therefore, $x + y = 90^\circ$, which means $\angle ACB$ is a right angle. $\square$



Sidenote: That any angle inscribed in a semicircle is a right angle is sometimes called **Thales Theorem**, after the mathematician **Thales of Miletus**, who is regarded by many as the father of the idea of 'proof'. He is credited by many with the first proofs that vertical angles are equal, that the sum of the angles in a triangle equals two right angles, and that the base angles of an isosceles triangle are equal.

Thales was an outstanding problem-solver. Legend has it that a farmer once complained to Thales, saying the farmer's donkey would lighten its load of salt on trips to the market by rolling in a river to dissolve the salt. Thales suggested the farmer load the donkey with sponges next time.
Source: *Journey Through Genius* by William Dunham

Important: Any angle inscribed in a semicircle is a right angle.



Having shown that any angle inscribed in a 180° arc has measure $180^\circ/2 = 90^\circ$, we should wonder if there's a general rule relating the measure of an inscribed angle to arcs in the circle.

Concept: When trying to prove something or find some general rule, investigating a few sample cases can be an excellent guide.



We'll try this strategy here.

Problem 12.2: Points A , B , and C are on $\odot O$ such that $\widehat{AC} = 80^\circ$ and $\widehat{ACB} = 130^\circ$. Find $\angle ABC$.

Solution for Problem 12.2: We know that the measure of an arc equals the angle formed by the radii that cut off the arc (we call such an angle a **central angle**). Therefore, we draw radii to A , B , and C , thus forming some isosceles triangles. Since $\angle BOC = \widehat{BC} = \widehat{AB} - \widehat{AC} = 50^\circ$, we have

$$\angle OBC = \angle OCB = \frac{180^\circ - 50^\circ}{2} = 65^\circ.$$

Similarly, $\angle AOB = \widehat{AB} = 130^\circ$, so

$$\angle OAB = \angle OBA = \frac{180^\circ - 130^\circ}{2} = 25^\circ.$$

Therefore, $\angle ABC = \angle OBC - \angle OBA = 40^\circ$.

We see that $\angle ABC = \widehat{AC}/2$ and we wonder if this is always the case. We can try changing $\widehat{BC}$ to see if that matters. If we let $\widehat{BC} = 64^\circ$, we can go through the same series of calculations as above to find that, indeed, $\angle ABC$ is still 40° . $\square$

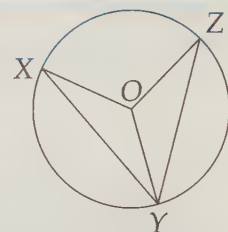
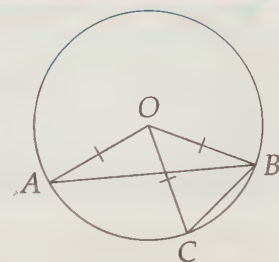
Now that we have a specific case as a guideline, we'll try to prove that an inscribed angle is always half the arc it intercepts. Unfortunately, to completely prove this, we'll need a number of cases. We'll try one of these cases here, then leave the others for Exercises.

Problem 12.3: Prove that if $\angle XYZ$ is inscribed in a circle such that the center of the circle, O , is inside $\triangle XYZ$, then $\angle XYZ = \widehat{XZ}/2$.

Solution for Problem 12.3: We use our specific case as a guide, and we connect the radii to X , Y , and Z to form isosceles triangles. We don't know any angles this time, so we'll have to use variables. Since we know we want to use our isosceles triangles, we let $\angle OYX = x$ and $\angle OYZ = z$, so that $\angle XYZ = x + z$. Isosceles triangles $\triangle XOY$ and $\triangle ZOY$ tell us that $\angle OXY = x$ and $\angle OZY = z$. Therefore, we have $\angle XOY = 180^\circ - 2x$ from $\triangle XOY$, and $\angle ZOY = 180^\circ - 2z$ from $\triangle ZOY$. We can now find $\angle XOZ$ in terms of x and z :

$$\angle XOZ = 360^\circ - \angle XOY - \angle ZOY = 2x + 2z = 2(x + z).$$

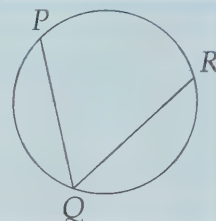
Therefore, $\widehat{XZ} = 2(x + z) = 2\angle XYZ$, so $\angle XYZ = \widehat{XZ}/2$. $\square$



Important:


The measure of an inscribed angle is one-half the measure of the arc it intercepts. For example,

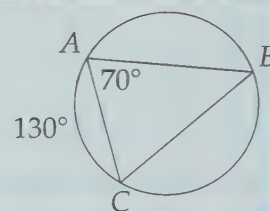
$$\angle PQR = \frac{\widehat{PR}}{2}.$$



Note that our proving that an angle inscribed in a semicircle is right is just a special case of this result. However, we draw special attention to the right angle case because right angles are so important in geometry.

Let's try using our newfound knowledge.

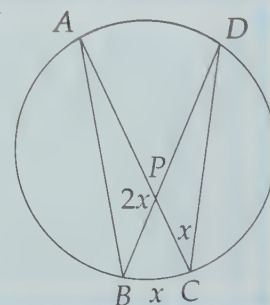
Problem 12.4: Given that $\triangle ABC$ is inscribed in the circle as shown, $\angle A = 70^\circ$, and $\widehat{AC} = 130^\circ$, find $\angle C$.



Solution for Problem 12.4: Solution 1: Since $\angle B$ is inscribed in $\widehat{AC}$, we have $\angle B = \widehat{AC}/2 = 65^\circ$. Therefore, $\angle C = 180^\circ - \angle A - \angle B = 45^\circ$.

Solution 2: Since $\angle A$ is inscribed in $\widehat{BC}$, we have $\widehat{BC} = 2\angle A = 140^\circ$. Therefore, we have $\widehat{AB} = 360^\circ - \widehat{AC} - \widehat{BC} = 90^\circ$. Since $\angle C$ is inscribed in $\widehat{AB}$, we have $\angle C = \widehat{AB}/2 = 45^\circ$. $\square$

Problem 12.5: Find x given that $\angle APB = 2x$, $\angle ACD = x$, and $\widehat{BC} = x$.



Solution for Problem 12.5: Since $\angle B$ and $\angle C$ are inscribed in the same arc, they must be equal (since each equals half the arc). Therefore, $\angle B = \angle C = x$. Since $\angle A$ is inscribed in $\widehat{BC}$, we have $\angle A = \widehat{BC}/2 = x/2$. Now we can use $\triangle APB$ to write an equation for x . Since

$$\angle A + \angle APB + \angle B = 180^\circ,$$

we have

$$\frac{x}{2} + 2x + x = 180^\circ.$$

Solving this equation gives $x = 51\frac{3}{7}^\circ$. $\square$

We could have solved the last problem in many different ways, but our solution above illustrates a

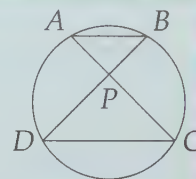
powerful principle that will be a crucial step in many problems when you move on to more advanced geometry.

Important: Any two angles that are inscribed in the same arc are equal.



We'll now use this important fact in a proof.

Problem 12.6: Chords $\overline{AC}$ and $\overline{BD}$ of a circle meet at P as shown. Given $AP = PB$, show that $\overline{AB} \parallel \overline{CD}$.



Solution for Problem 12.6: Since $AP = PB$, we have $\angle PBA = \angle PAB$. Since $\angle CAB$ and $\angle CDB$ are inscribed in the same arc, we have $\angle CDB = \angle CAB$. $\angle CAB$ is the same angle as $\angle PAB$, so we have $\angle CDB = \angle PAB = \angle PBA = \angle DBA$. Since $\angle CDB = \angle DBA$, we have $\overline{AB} \parallel \overline{CD}$. $\square$

Exercises

In Exercises 12.1.5 and 12.1.6 below, do not assume that an inscribed angle equals half the arc it intercepts. You are asked to prove this fact for various cases in these two problems.

12.1.1 Points P , Q , and R are on $\odot O$ such that $\widehat{PQ} = 78^\circ$, $\widehat{QR} = 123^\circ$, and $\widehat{PQR} = 201^\circ$.

- Find $\angle QPR$.
- Find $\angle PQR$.
- Find $\angle PRQ$.
- Find $\angle POQ$.
- Find $\angle PQO$.
- Find $\angle POR$. (Be careful on this one!)

12.1.2 Points R and S are on $\odot E$ such that $\widehat{RS} = 50^\circ$. Point T is also on $\odot E$. Find all possible values of $\angle RTS$.

12.1.3 Points E , F and G are on $\odot O$ such that $\angle EFG = 48^\circ$ and $\angle GEF = 78^\circ$.

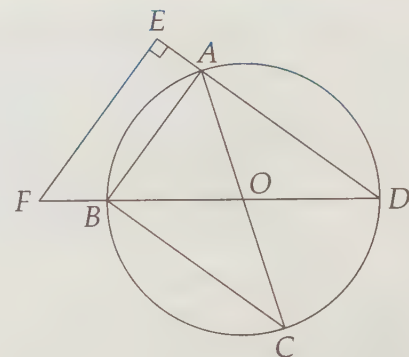
- Find $\angle EGF$.
- Find $\widehat{EG}$.
- Find $\widehat{EFG}$.

Extra! The advancement and perfection of mathematics are intimately connected with the prosperity of the state.

—Napoleon Bonaparte

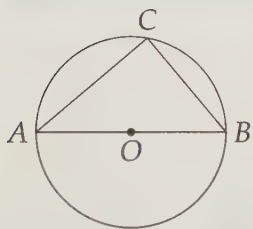
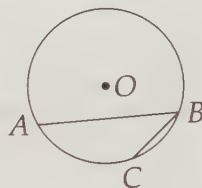
12.1.4 In the diagram, O is the center of the circle and $\angle ACB = 36^\circ$.

- Find $\angle D$.
- Find $\widehat{AB}$.
- Find $\angle ABF$. (Do not assume $\overline{EF} \parallel \overline{AB}$ at this point.)
- Find $\angle EFB$. (Do not assume $\overline{EF} \parallel \overline{AB}$ at this point.)
- Find $\widehat{CD}$.
- Prove that $\overline{EF} \parallel \overline{AB}$.
- Prove that $AD = BC$.



12.1.5 In Problem 12.2, we computed $\angle ABC$ in the diagram at right for specific measures of $\widehat{AC}$ and $\widehat{BC}$. We found that $\angle ABC = \widehat{AC}/2$.

- Prove that $\angle ABC = \widehat{AC}/2$ for all inscribed angles for which the center of the circle is outside the angle as shown in the figure at right. (Make sure you see how this problem is a little different from Problem 12.3.)
- Prove that $\angle ACB$ has measure equal to one-half major arc $\widehat{AB}$.



12.1.6 In Problem 12.1, we showed that an angle inscribed in a semicircle is a right angle. In this problem, we tackle the other angles that are formed when we connect the endpoints of a diameter of a circle to a point on the circle. $\overline{AB}$ is a diameter of $\odot O$ as shown. Prove that $\angle CAB = \widehat{BC}/2$. **Hints:** 83

12.1.7 Arcs $\widehat{WX}$ and $\widehat{YZ}$ of $\odot Q$ are congruent. Prove that either $\overline{WY} \parallel \overline{XZ}$ or $\overline{WZ} \parallel \overline{XY}$.

12.1.8 Back in Problem 9.5 of Chapter 9, we considered the regular 15-gon $ABCDEFGHIJKLMNO$, and found $\angle ACD$ and $\angle ADE$. Find another solution with your new knowledge about angles and circles, by considering the circumcircle of the polygon to find $\angle ACD$ and $\angle ADE$.

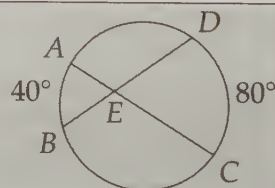
12.2 Angles Inside and Outside Circles

A line that intersects a circle in two points is called a **secant**. In this chapter we'll explore relationships between the angles formed by intersecting secants and the arcs of circles these angles cut off.

Problems

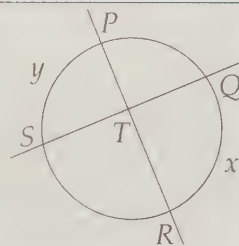
Problem 12.7: Our goal in this problem is to find $\angle AEB$ given that $\widehat{AB} = 40^\circ$ and $\widehat{DC} = 80^\circ$.

- Draw $\overline{AD}$ to create inscribed angles. Find $\angle DAE$ and $\angle ADE$.
- Find $\angle AED$ and $\angle AEB$.
- How is $\angle AEB$ related to $\widehat{AB}$ and $\widehat{CD}$?



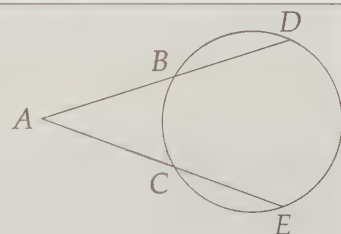
Problem 12.8: In this problem we will find a relationship between an angle inside a circle and the arcs the angle intercepts. In the diagram, let $\widehat{QR} = x$ and $\widehat{PS} = y$.

- Draw $\overline{PQ}$ and find $\angle PQT$ and $\angle QPT$ in terms of x and/or y .
- Find, with proof, $\angle QTR$ in terms of x and y .

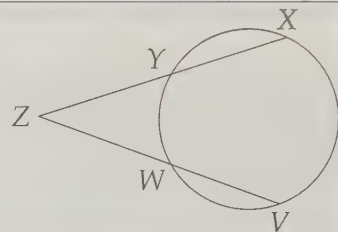


Problem 12.9: In this problem we will find $\angle DAE$ given $\widehat{DE} = 110^\circ$ and $\widehat{BC} = 20^\circ$.

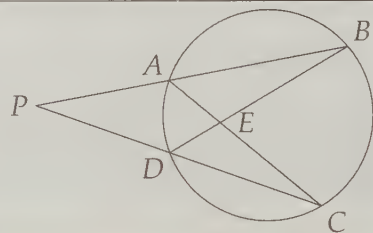
- Draw $\overline{BE}$ to create inscribed angles. Find $\angle BEC$ and $\angle DBE$.
- Find $\angle DAE$.
- Can you find a general relationship that must hold among $\angle A$, $\widehat{BC}$, and $\widehat{DE}$? (In other words, what if we replace 110° and 20° with x and y ?)



Problem 12.10: Given that $\widehat{XV} = b$ and $\widehat{YW} = a$ (these are arc measures, not lengths), prove that $\angle Z = (b - a)/2$.

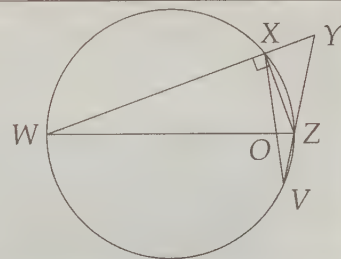


Problem 12.11: Given that $\widehat{AD} = 40^\circ$, $\widehat{AB} = 120^\circ$, and $\widehat{DC} = 100^\circ$, find $\angle BPC$ and $\angle BEC$.



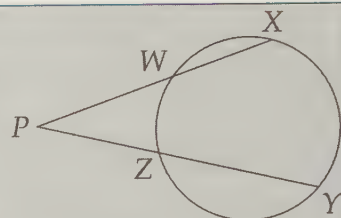
Problem 12.12: In the figure, $\angle Y = 58^\circ$, $\angle W = 20^\circ$, and $\overline{XZ} \perp \overline{WX}$.

- Find $\angle WOV$.
- Find $\angle ZXV$.



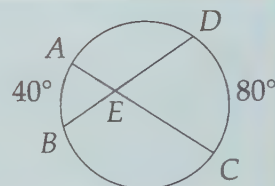
Extra! You'll sometimes see the letters 'Q.E.D.' at the end of a mathematical proof. These letters are an abbreviation for the Latin phrase *quod erat demonstratum*, which means 'which was to be proved.'

Problem 12.13: $\angle P$ meets the circle at four points as shown. The arcs $\widehat{WX}$, $\widehat{WZ}$, $\widehat{XY}$, and $\widehat{YZ}$ are in order from smallest to largest, and each of the last three is 42° larger than the arc listed before it. Find $\angle P$. **Hints:** 23 (Source: Mandelbrot)



Having found a general formula for inscribed angles, we wonder how angles formed by secants are related to the arcs they intercept. We start, of course, with a specific example.

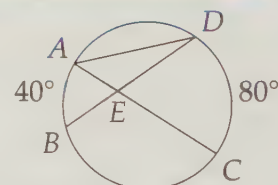
Problem 12.7: Find $\angle AEB$ given that $\widehat{AB} = 40^\circ$ and $\widehat{DC} = 80^\circ$.



Solution for Problem 12.7: We try to put what we just learned about inscribed angles to work by drawing $\overline{AD}$ to create some inscribed angles. We have $\angle DAC = \widehat{CD}/2 = 40^\circ$ and $\angle ADB = \widehat{AB}/2 = 20^\circ$. Therefore,

$$\angle AED = 180^\circ - \angle EAD - \angle EDA = 120^\circ,$$

so, $\angle AEB = 180^\circ - \angle AED = 60^\circ$. (Note that we could have seen that $\angle AEB = 20^\circ + 40^\circ$ since it is an exterior angle of $\triangle AED$.) $\square$



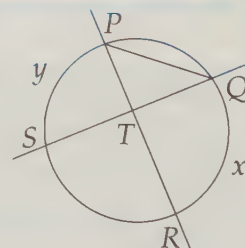
With that basic case as a guide, we're ready to tackle finding a general formula.

Problem 12.8: Chords $\overline{QS}$ and $\overline{PR}$ intersect at T . Given that $\widehat{QR} = x$ and $\widehat{PS} = y$, find a formula for $\angle QTR$ in terms of x and y .

Solution for Problem 12.8: We form inscribed angles by adding $\overline{PQ}$ to our diagram. Since $\angle QPR$ is inscribed in $\widehat{QR}$, $\angle QPR = x/2$. Similarly, $\angle PQS = y/2$. Since $\angle QTR$ is an exterior angle of $\triangle PQT$, we have

$$\angle QTR = \angle PQT + \angle QPT = \frac{x+y}{2}.$$

$\square$

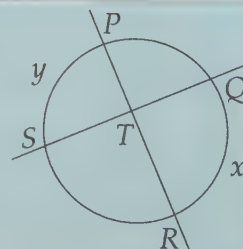


Important:



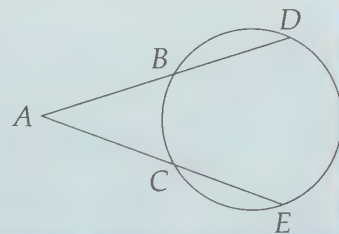
The measure of the angle formed by two intersecting chords is the average of the measures of the arcs intersected by the chords. For example, in the diagram at right, we have

$$\angle PTS = \angle QTR = \frac{\widehat{PS} + \widehat{QR}}{2} = \frac{x+y}{2}.$$



We've tackled angles on a circle and angles inside a circle. You know where we're headed now: outside the circle.

Problem 12.9: Find $\angle DAE$ given $\widehat{DE} = 110^\circ$ and $\widehat{BC} = 20^\circ$.



Solution for Problem 12.9: We succeeded with inscribed angles before, so we try using them again here by drawing $\overline{BE}$. We have $\angle BEC = 20^\circ/2 = 10^\circ$ and $\angle DBE = 110^\circ/2 = 55^\circ$. Thus, $\angle ABE = 180^\circ - \angle DBE = 125^\circ$, and we have

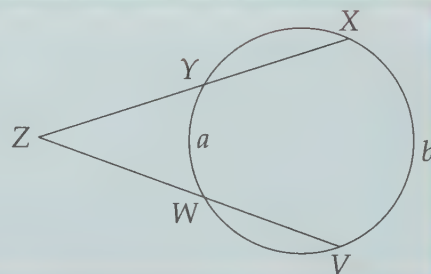
$$\angle DAE = 180^\circ - \angle ABE - \angle BEC = 180^\circ - 125^\circ - 10^\circ = 45^\circ.$$

Note that we could have seen that $\angle DBE$ is an exterior angle of $\triangle ABE$, so $\angle DBE = \angle BAE + \angle BEC$. As before, this gives us $\angle BAE = \angle DBE - \angle BEC = 45^\circ$. $\square$

Now we have a clear path to prove the general formula.

Problem 12.10: Given that $\widehat{XV} = b$ and $\widehat{YW} = a$ (these are arc measures, not lengths) in the diagram, prove that

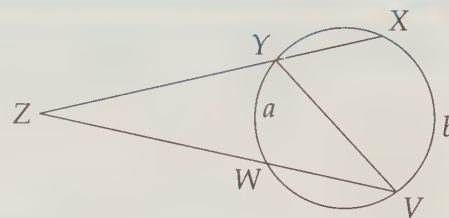
$$\angle Z = \frac{b - a}{2}.$$



Solution for Problem 12.10: We draw $\overline{YV}$ and have $\angle YVW = a/2$ and $\angle XYV = b/2$. Since $\angle XYV$ is an exterior angle of $\triangle VYZ$, we have $\angle XYV = \angle Z + \angle YVZ$. Therefore,

$$\angle Z = \angle XYV - \angle YVZ = \frac{b - a}{2}.$$

$\square$

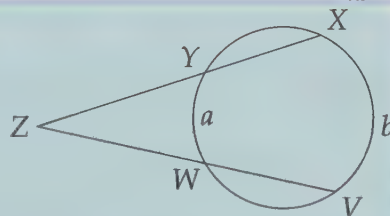


Important:



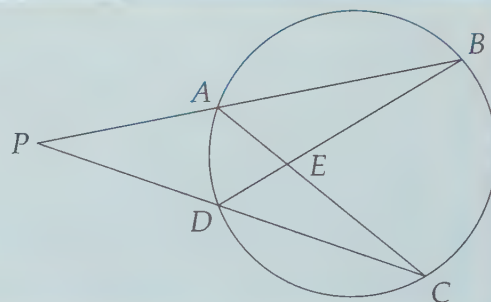
Two secants that meet at a point outside a circle form an angle equal to half the difference of the arcs they intercept. For example, in the diagram we have

$$\angle Z = \frac{b - a}{2}.$$



Let's solve a few problems with our new relationships.

Problem 12.11: Given that $\widehat{AD} = 40^\circ$, $\widehat{AB} = 120^\circ$, and $\widehat{DC} = 100^\circ$, find $\angle BPC$ and $\angle BEC$.



Solution for Problem 12.11: Since all the arcs together in a circle must sum to 360° , we have $\widehat{BC} = 360^\circ - 120^\circ - 40^\circ - 100^\circ = 100^\circ$. Therefore, we have

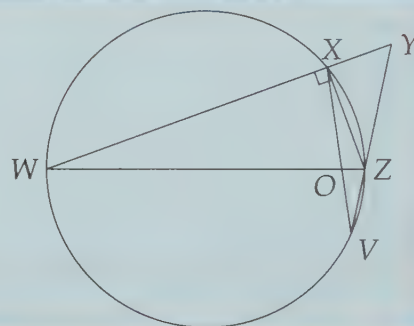
$$\begin{aligned}\angle BEC &= \frac{\widehat{BC} + \widehat{AD}}{2} = 70^\circ \\ \angle BPC &= \frac{\widehat{BC} - \widehat{AD}}{2} = 30^\circ\end{aligned}$$

□

Not all problems can be solved in one or two steps. Here's an example in which we have to do a little more work.

Problem 12.12: In the figure, $\angle Y = 58^\circ$, $\angle W = 20^\circ$, and $\overline{XZ} \perp \overline{WX}$.

- Find $\angle WOV$.
- Find $\angle ZXV$.

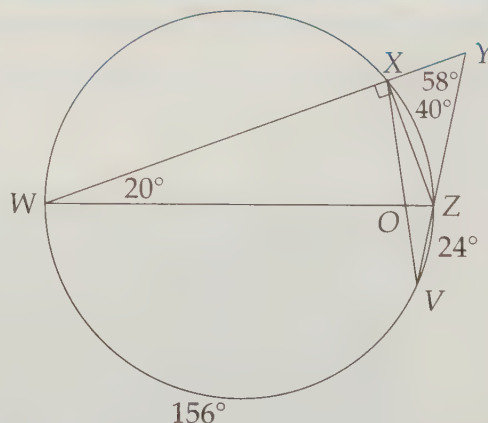


Solution for Problem 12.12: We can't find the angles we want right away, but we have a circle with our angles, so we go hunting for measures of arcs. Since $\angle XWZ = 20^\circ$, we know that $\widehat{XZ} = 40^\circ$. Since $\angle Y = (\widehat{WV} - \widehat{XZ})/2$, we have

$$58^\circ = \frac{\widehat{WV} - 40^\circ}{2}.$$

Solving for $\widehat{WV}$ gives $\widehat{WV} = 156^\circ$. Now we can find $\angle WOV$:

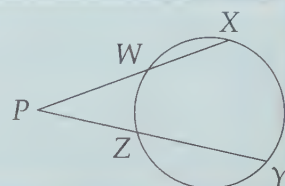
$$\angle WOV = \frac{\widehat{WV} + \widehat{XZ}}{2} = \frac{156^\circ + 40^\circ}{2} = 98^\circ.$$



We can use $\widehat{VZ}$ to find $\angle ZXV$, so we focus on $\widehat{VZ}$. Since $\widehat{WVZ} = 2\angle ZXW = 180^\circ$, we have $\widehat{VZ} = 180^\circ - \widehat{WV} = 24^\circ$. Therefore, $\angle ZXV = 12^\circ$. □

Concept: Exploration can help unravel problems whose solutions require several steps. In geometry problems, effective exploration often requires drawing an accurate diagram and finding any angle measures or segment lengths you can. When you find any new information, you should label your diagram with your new information, as we did in the solution to Problem 12.12. This helps you see when you have the information you need to solve the problem.

Problem 12.13: $\angle P$ meets the circle at four points as shown. The arcs $\widehat{WX}$, $\widehat{WZ}$, $\widehat{XY}$, and $\widehat{YZ}$ are in order from smallest to largest, and each of the last three is 42° larger than the arc listed before it. Find $\angle P$. (Source: Mandelbrot)

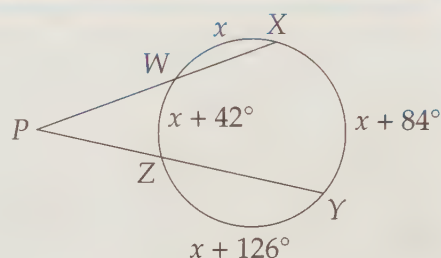


Solution for Problem 12.13: We start by labeling our diagram with the given information. We let our smallest arc have measure x , and we are given that the other three arcs are $x + 42^\circ$, $x + 84^\circ$, and $x + 126^\circ$. We could find x if we wanted to, but we don't have to, because

$$\angle P = \frac{\widehat{XY} - \widehat{WZ}}{2} = \frac{(x + 84^\circ) - (x + 42^\circ)}{2} = 21^\circ.$$

□

Just because we use a variable in a problem doesn't mean we have to find its value. However, the variable did make finding the solution easier.



Concept: When you can't find any more lengths or angles in a problem but you still haven't solved it, try assigning a variable to one of the lengths or angles. Then, find other lengths or angles in terms of that variable. Finally, label your diagram with everything you find – this will make it easier to see when you have enough information to solve the problem.

Exercises

12.2.1 In the diagram at left below, $\widehat{AB} = 40^\circ$, $\widehat{BC} = 103^\circ$, and $\widehat{CD} = 83^\circ$. Find $\angle AED$.

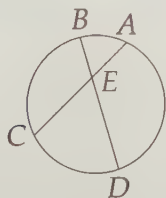


Figure 12.2: Diagram for Problem 12.2.1

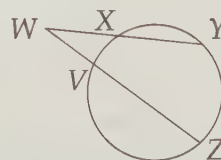


Figure 12.3: Diagram for Problem 12.2.2

12.2.2 Find $\widehat{YZ}$ and $\widehat{VZ}$ in the diagram at right above if $\widehat{XY} = 78^\circ$, $\widehat{XV} = 32^\circ$, and $\angle YWZ = 30.5^\circ$.

12.2.3 Points P , Q , R , and S are in that order on $\odot O$. $\overline{PR}$ and $\overline{QS}$ meet at Z . Given that $\widehat{PQ} = 2\widehat{QR} = 3\widehat{RS} = 4\widehat{SP}$, find $\angle QZR$. **Hints:** 336

12.2.4 Given that $\widehat{XY} = 75^\circ$, $\widehat{XYZ} = 117^\circ$, and $\widehat{YZW} = 173^\circ$ in the diagram at left below, find the following:

- (a) $\angle YPZ$
- (b) $\angle YOZ$
- (c) $\angle XYW$
- (d) $\angle OWZ$

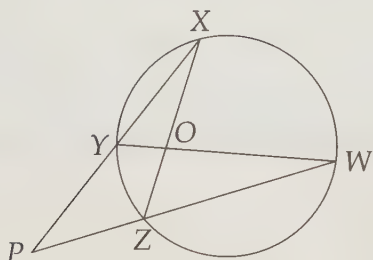


Figure 12.4: Diagram for Problem 12.2.4

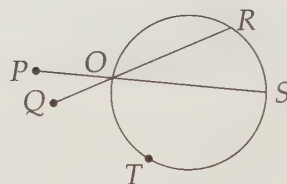


Figure 12.5: Diagram for Problem 12.2.5

12.2.5 Prove that $\angle POR = \frac{\widehat{OR} + \widehat{OTS}}{2}$ in the diagram at right above. **Hints:** 433

12.3 Tangents

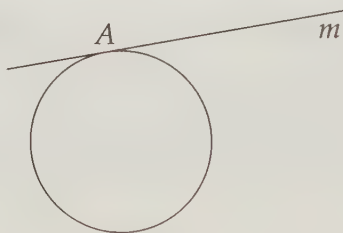
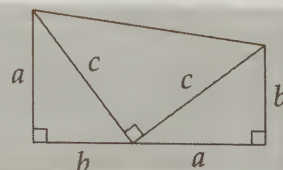


Figure 12.6: A Tangent Line

A line is **tangent** to a circle when it touches the circle in only one point. For example, line m is tangent to the circle at point A in the diagram above.

In this section we explore how a tangent line is related to the circle it touches.

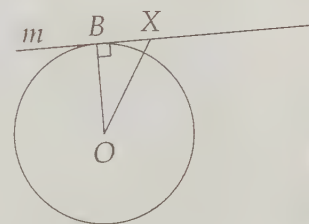
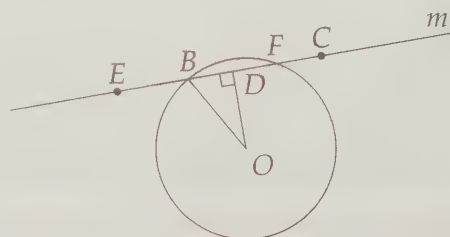
Extra! President James A. Garfield used the diagram at right to prove the Pythagorean Theorem. How did he do it? **Hints:** 5



Problems

Problem 12.14: Line m passes through point B on a circle with center O . In this problem, we will first show that if m is tangent to the circle, then $m \perp \overline{OB}$. We do so by showing that if $\angle OBC$ is not 90° , then m must hit the circle a second time. Then, we will show that if $m \perp \overline{OB}$, then m must be tangent to the circle.

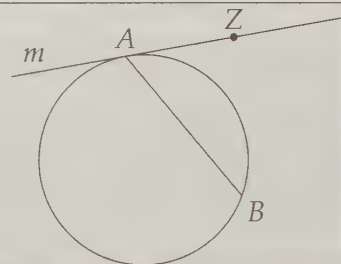
- Suppose $\angle OBC < 90^\circ$ as shown at right. If so, there must be some other line through O besides $\overleftrightarrow{OB}$ that is perpendicular to m . Let this line hit m at point D . What kind of triangle is $\triangle OBD$?
- Show that $OD < OB$. Why does this mean that m must hit the circle a second time?
- Show that if $\angle OBC > 90^\circ$, then m must hit the circle a second time. Conclude that if m is tangent to the circle at B , then $m \perp \overline{OB}$.
- Our first three parts show that a tangent line must be perpendicular to the radius drawn to the point of tangency. Now, we show that if a line m passes through a point B on a $\odot O$ such that $m \perp \overline{OB}$, then m must be tangent to the circle.



Suppose line m meets $\odot O$ at B such that $m \perp \overline{OB}$. Let X be any point on m besides B . Consider $\triangle OBX$ and show that $OX > OB$. Does this prove that m can't hit the circle a second time if $m \perp \overline{OB}$?

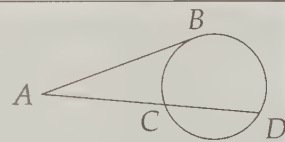
Problem 12.15: Our goal in this problem is to find $\angle ZAB$ given that line m is tangent to the circle at A and $\widehat{AB} = 80^\circ$.

- Draw diameter $\overline{AC}$ and find $\widehat{BC}$.
- Find $\angle BAC$, $\angle ZAC$, and $\angle ZAB$.
- What relationship holds between $\angle ZAB$ and $\widehat{AB}$?



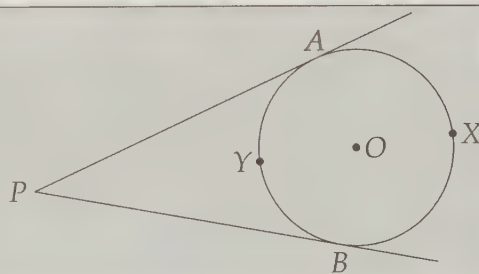
Problem 12.16: In this problem we will explore how an angle between a secant and a tangent is related to the arcs intercepted by the angle. In the diagram, let $\widehat{BD} = 110^\circ$ and $\widehat{BC} = 40^\circ$. $\overline{AB}$ is tangent to the circle at B .

- Draw $\overline{BD}$ and find $\angle BDC$ and $\angle DBA$.
- Find $\angle A$.



Problem 12.17: In this problem we investigate the relationship between the lengths of two tangent segments to a circle from the same point, as well as how to find the angle between them. $\overrightarrow{PA}$ and $\overrightarrow{PB}$ are tangent to the circle, which has center O .

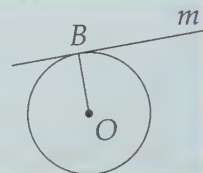
- Prove that $\triangle POB \cong \triangle POA$.
- Prove $\angle P = \frac{\widehat{AXB} - \widehat{AYB}}{2}$.



Problem 12.18: Point P is 25 centimeters from the center, O , of a circle with radius 10 centimeters. Point Q is on the circle such that $\overline{PQ}$ is tangent to the circle. Find PQ .

We start this section discovering a highly useful fact about tangents.

Problem 12.14: Given that line m is tangent to $\odot O$ at B , prove that radius $\overline{OB}$ is perpendicular to m . Conversely, prove that if a line m passes through point B on $\odot O$ such that $\overline{OB} \perp m$, then m is tangent to $\odot O$ at B .



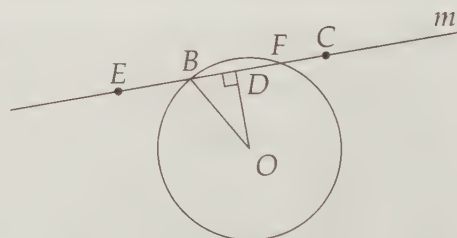
Solution for Problem 12.14: This is one of those ‘obvious’ facts in geometry that isn’t so easy to prove rigorously. Here’s one ‘obvious’ solution that doesn’t quite work:

Bogus Solution: Since the diagram is the same on both sides of $\overline{OB}$, the angles on both sides of B must be the same. These angles must also add to 180° , so they must be right angles.



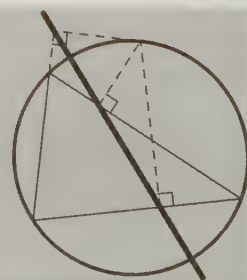
The problem with this solution is that the statement ‘the diagram is the same on both sides of $\overline{OB}$ ’ essentially assumes what we are trying to prove. Moreover, the solution is not clearly written – what do we mean by ‘the angles on both sides of B ’? We’ll have to find another way.

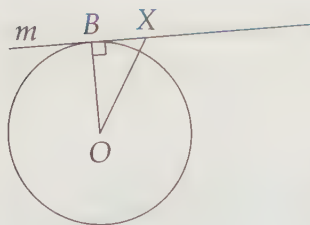
We let C be a point on m besides B . After trying and failing to directly show that $\angle OBC = 90^\circ$, we decide to try proving it can’t be anything else. First, we consider $\angle OBC < 90^\circ$. Since $\angle OBC$ is not 90° , there must be some other point D on line m such that $\overline{OD} \perp m$. Hypotenuse $\overline{OB}$ of right triangle $\triangle OBD$ must be the largest side of $\triangle OBD$, so $OD < OB$. Since $\overline{OB}$ is a radius, this means that line m must go inside the circle if $\angle OBC < 90^\circ$. Therefore, the line will hit the circle a second time. Specifically, consider the point F on m such that $BD = DF$ (i.e., D is the midpoint of $\overline{BF}$). By SAS Congruence we have $\triangle BDO \cong \triangle FDO$, so $OB = OF$. Therefore, point F is on $\odot O$, so if $\angle OBC < 90^\circ$, then line m cannot be tangent to the circle since it will hit the circle twice.



Similarly, if $\angle OBC > 90^\circ$, then $\angle OBE < 90^\circ$, and we can use exactly the same logic to show that m cannot be a tangent line in this case either. Therefore, if m does not form a 90° angle with $\overline{OB}$, it cannot be a tangent line.

Extra! The feet of the perpendiculars from any point on the circumcircle of a triangle to the sides of the triangle are collinear, as shown at right. The line through these points is called the **Simson line**. See if you can prove that these three points are always collinear! **Hints:** 21





Unfortunately, we are not finished yet! We have shown that a line that makes any angle besides 90° can't be a tangent line, but we haven't shown that a line that makes a 90° angle is always a tangent. It might even be true that it's impossible for there to ever be a tangent line. Fortunately, it's not, and we'll prove it by considering line m at left, which hits the circle at B such that $m \perp \overline{OB}$. To show that m is a tangent line, we must show that it does not hit the circle again.

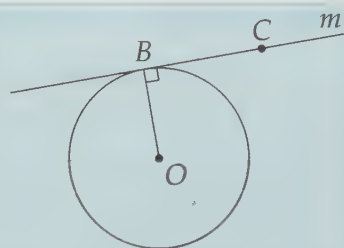
We consider some point on line m besides point B , and call this point X . Since $\overline{OX}$ is the hypotenuse of $\triangle OBX$, we must have $OX > OB$. Hence, point X must be outside the circle, since its distance from O is greater than the circle's radius. This is true for any point on line m besides point B , so all points on m besides B are outside the circle. Thus, line m is indeed tangent to the circle at point B . $\square$

We therefore have a very non-obvious proof for our 'obvious' fact that:

Important:



A line tangent to a circle is perpendicular to the radius drawn to the point of tangency. Conversely, a line drawn through a point on a circle that is perpendicular to the radius drawn to that point must be tangent to the circle. For example, for the diagram at right, we can write:



$\overleftrightarrow{BC}$ is perpendicular to radius $\overline{OB}$ if and only if $\overleftrightarrow{BC}$ is tangent to circle O .

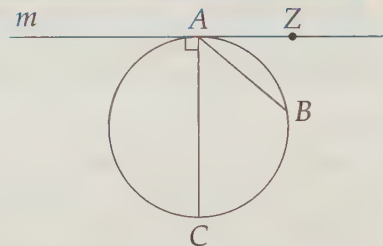
You'll be seeing this 'radius perpendicular to a tangent at the point of tangency' again and again.

Problem 12.15: Line m is tangent to a circle at point A . Given point B on the circle such that $\widehat{AB} = 80^\circ$, find the acute angle formed by chord $\overline{AB}$ and line m .

Solution for Problem 12.15: We start by drawing diameter $\overline{AC}$, since we know this line is perpendicular to m at point A . Since $\overline{AC}$ is a semicircle, $\widehat{BC} = 180^\circ - \widehat{AB} = 100^\circ$. Thus, $\angle BAC = \widehat{BC}/2 = 50^\circ$. Finally, $\angle ZAB = \angle ZAC - \angle BAC = 90^\circ - 50^\circ = 40^\circ$.

Note that $\angle ZAB = \widehat{AB}/2$. $\square$

As an Exercise, you can follow these same steps to show:

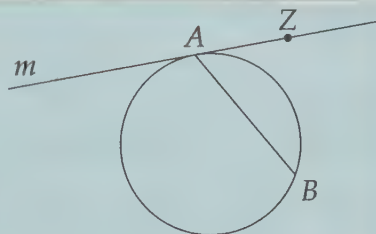


Important:



An angle formed by a tangent and a chord with the point of tangency as an endpoint equals one-half the arc intercepted by the angle. For example, in the figure at right, line m is tangent to the circle at A , so

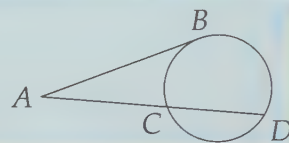
$$\angle ZAB = \frac{\widehat{AB}}{2}.$$



An angle between a tangent and a chord is also often referred to as an **inscribed angle**. Yes, this is the same name we give to an angle between two chords that share an endpoint; the measures of both types of angles equal half the measures of the arcs they cut off.

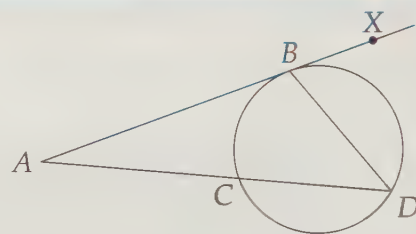
We now turn to angles formed by a tangent and a secant.

Problem 12.16: In the diagram, $\widehat{BD} = 110^\circ$ and $\widehat{BC} = 40^\circ$. $\overleftrightarrow{AB}$ is tangent to the circle at B . Find $\angle BAD$.



Solution for Problem 12.16: Since this looks so much like problems involving an angle between two secants (such as Problem 12.9), we try the same approach by drawing $\overline{BD}$. We have $\angle BDC = \widehat{BC}/2 = 20^\circ$ and $\angle XBD = \widehat{BD}/2 = 55^\circ$.

Finally, since $\angle XBD$ is an exterior angle of $\triangle ABD$, we have $\angle XBD = \angle BAD + \angle BDC$, so $\angle BAD = \angle XBD - \angle BDC = 55^\circ - 20^\circ = 35^\circ$. $\square$

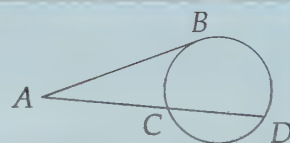


Unsurprisingly, the angle formed by a tangent and a secant has the same relationship to the arcs the angle intercepts as an angle formed by two secants does.

Important: The angle formed by a tangent and a secant is half the difference of the intercepted arcs. For example,



$$\angle BAD = \frac{\widehat{BD} - \widehat{BC}}{2}.$$



Next we consider two tangents from the same point to the same circle.

Problem 12.17: Segments $\overline{PA}$ and $\overline{PB}$ are tangent to the same circle at A and B , respectively. Prove that $PA = PB$, and that $\angle APB$ equals half the difference of major arc $\widehat{AB}$ and minor arc $\widehat{AB}$.

Solution for Problem 12.17: We start by drawing the radii to the points of tangency, thus forming right angles. We also draw $\overline{PO}$ to complete a pair of right triangles. Since $AO = OB$, we have $\triangle OAP \cong \triangle OBP$ by HL Congruence. Therefore, $PA = PB$.

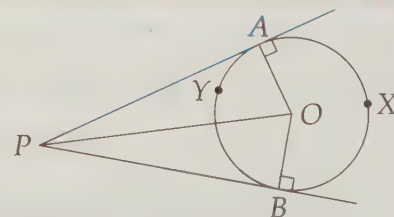
We can find $\angle APB$ a variety of ways. One is to use quadrilateral $PAOB$ to note that

$$\angle APB = 360^\circ - \angle PAO - \angle AOB - \angle OBP = 180^\circ - \angle AOB = 180^\circ - \widehat{AYB}.$$

Since $\widehat{AXB} = 360^\circ - \widehat{AYB}$, we also have

$$\frac{\widehat{AXB} - \widehat{AYB}}{2} = \frac{360^\circ - 2\widehat{AYB}}{2} = 180^\circ - \widehat{AYB} = \angle APB.$$

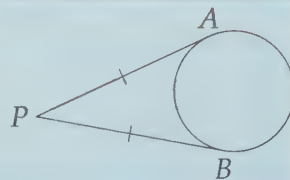
$\square$



The information about the angle between the tangents is unsurprising, as it's essentially the same as the angle between two secants. However, the information about the two tangents from the same point is new, and is one of the most useful tangent facts:

Important:

As shown in the diagram at right, we can draw two tangent segments to a circle from a point outside the circle. These tangents are always equal to each other in length.



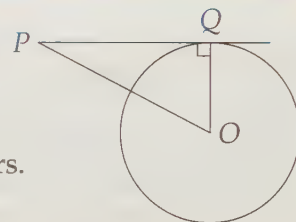
We'll finish with a very common problem involving tangents.

Problem 12.18: Point P is 25 centimeters from the center, O , of a circle with radius 10 centimeters. Point Q is on the circle such that $\overline{PQ}$ is tangent to the circle. Find PQ .

Solution for Problem 12.18: We start by drawing a radius to the point of tangency as shown. Since $\overline{PQ}$ is tangent to the circle, $\triangle POQ$ is a right triangle. Thus,

$$OP^2 = OQ^2 + PQ^2, \text{ so}$$

$$PQ = \sqrt{OP^2 - OQ^2} = \sqrt{25^2 - 10^2} = \sqrt{5^2(5^2 - 2^2)} = 5\sqrt{5^2 - 2^2} = 5\sqrt{21} \text{ centimeters.}$$



Notice how a little factoring makes our arithmetic easier. $\square$

Exercises

12.3.1 Points U and I are on $\odot O$ such that major arc $\widehat{UI}$ is twice as long as minor arc $\widehat{UI}$. P is outside the circle such that $\overline{PU}$ and $\overline{PI}$ are tangent to $\odot O$.

- Find minor arc $\widehat{UI}$.
- Find $\angle UPI$.
- Find $\angle PIO$.
- Find $\angle IOU$.
- Find $\angle PIU$.

12.3.2 Point T is outside $\odot O$. X is on $\odot O$ such that $\overline{TX}$ is tangent to $\odot O$. The radius of $\odot O$ is 6 and $TX = 12$. Find TO .

12.3.3 Generalize Problem 12.15 by proving that the angle formed by a tangent and a chord that has the point of tangency as an endpoint is half the arc intercepted by the angle.

12.3.4 Y and Z are on $\odot G$ and X is outside $\odot G$ such that $\overline{XY}$ and $\overline{XZ}$ are tangent to $\odot G$. Given that $\angle YXZ = 51^\circ$, find the measure of minor arc $\widehat{YZ}$.

12.3.5 Given a circle and a point outside the circle, it is intuitively clear that there are exactly two lines through the point that are tangent to the circle. However, intuitively isn't good enough – we're going to prove it. Let the point be P and the circle be $\odot O$.

- (a) Consider the circle with diameter $\overline{OP}$. Call this circle C . Why must C hit $\odot O$ in at least two different points?
- (b) Why is it impossible for C to hit $\odot O$ in *three* different points? **Hints:** 530
- (c) Let the points where C hits $\odot O$ be A and B . Prove that $\angle PAO = \angle PBO = 90^\circ$.
- (d) Prove that $\overline{PA}$ and $\overline{PB}$ are tangent to $\odot O$.
- (e)★ Now for the tricky part – proving that these are the only two tangents. Suppose D is on $\odot O$ such that $\overline{PD}$ is tangent to $\odot O$. Why must D be on C ? **Hints:** 478
- (f) Why does the previous part tell us that $\overrightarrow{PA}$ and $\overrightarrow{PB}$ are the only lines through P tangent to $\odot O$?

12.3.6★ $\odot O$ is tangent to all four sides of rhombus $ABCD$, $AC = 24$, and $AB = 15$.

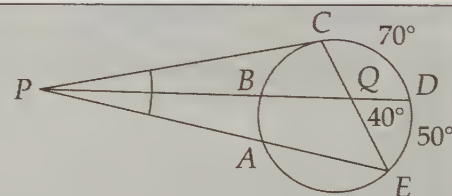
- (a) Prove that $\overline{AC}$ and $\overline{BD}$ meet at O (i.e., prove that the intersection of the diagonals of $ABCD$ is the center of the circle.) **Hints:** 439
- (b) What is the area of $\odot O$? **Hints:** 508

12.4 Problems

Problems

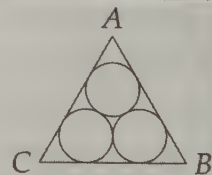
Problem 12.19: A quadrilateral is said to be a **cyclic quadrilateral** if a circle can be drawn that passes through all four of its vertices. Prove that if $ABCD$ is a cyclic quadrilateral, then $\angle A + \angle C = 180^\circ$.

Problem 12.20: In the figure, $\overline{PC}$ is tangent to the circle and $\overline{PD}$ bisects $\angle CPE$. Furthermore, $\widehat{CD} = 70^\circ$, $\widehat{DE} = 50^\circ$, and $\angle DQE = 40^\circ$. In this problem we determine the measure of the arc from A to E that does not include point C .

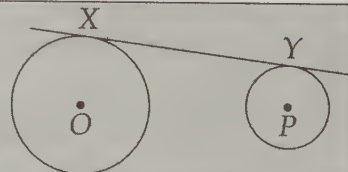


- (a) Find $\widehat{BC}$.
- (b) Find $\angle CPB$.
- (c) Find $\widehat{AB}$.
- (d) Finish the problem.

Problem 12.21: Three congruent circles with radius 1 are drawn inside equilateral $\triangle ABC$ such that each circle is tangent to the other two and to two sides of the triangle. Find the length of a side of $\triangle ABC$. **Hints:** 428



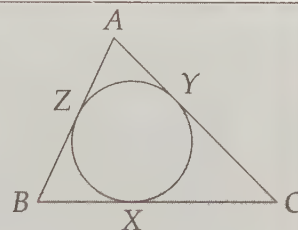
Problem 12.22: $\overleftrightarrow{XY}$ is tangent to both circle O and circle P . Given that $OP = 40$, and the radii of circles O and P are 31 and 7, respectively, find XY . **Hints:** 548, 267



Problem 12.23: Median $\overline{AM}$ of $\triangle ABC$ has length 8. Given that $BC = 16$ and $AB = 9$, find the area of $\triangle ABC$. **Hints:** 25, 187

Problem 12.24: The diagram shows the incircle of $\triangle ABC$. X , Y , and Z are the points of tangency where the incircle touches the triangle. In this problem we will find an expression for AZ in terms of the sides of the triangle.

- Find equal segments in the diagram and assign them variables.
- Let $AB = c$, $AC = b$, and $BC = a$. Use your variables from the first part to write equations that include these lengths.
- Solve your resulting equations for AZ . **Hints:** 157



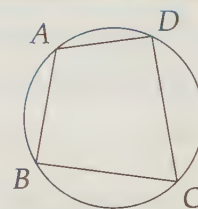
We'll now apply our knowledge of circles, angles, and tangents to more challenging problems and develop some useful geometric concepts.

Problem 12.19: A quadrilateral is said to be a **cyclic quadrilateral** if a circle can be drawn that passes through all four of its vertices. Prove that if $ABCD$ is a cyclic quadrilateral, then $\angle A + \angle C = 180^\circ$.

Solution for Problem 12.19: Since $\angle A$ and $\angle C$ are inscribed angles, we have

$$\angle A + \angle C = \frac{\widehat{BCD}}{2} + \frac{\widehat{BAD}}{2} = \frac{\widehat{BCD} + \widehat{BAD}}{2} = \frac{360^\circ}{2} = 180^\circ.$$

□

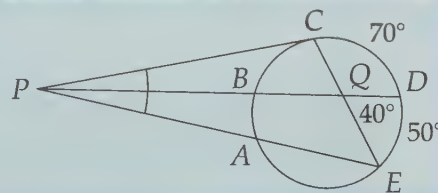


Important: A quadrilateral is a **cyclic quadrilateral** if a circle can be drawn that passes through all four of its vertices. Such a quadrilateral is said to be **inscribed** in the circle. The opposite angles of any cyclic quadrilateral sum to 180° .



You'll be seeing a lot more of cyclic quadrilaterals when you move into more advanced geometry.

Problem 12.20: In the figure, $\overline{PC}$ is tangent to the circle and $\overline{PD}$ bisects $\angle CPE$. If $\widehat{CD} = 70^\circ$, $\widehat{DE} = 50^\circ$, and $\angle DQE = 40^\circ$, then determine the measure of the arc from A to E that does not include point C .



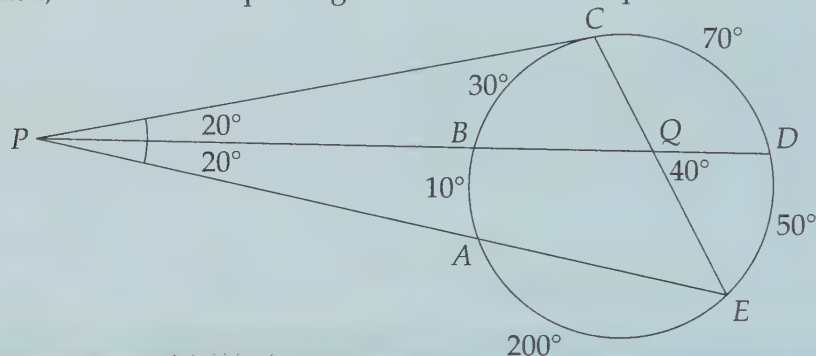
Solution for Problem 12.20: We can't directly find the desired arc, so we try finding whatever we can. First, we note that since $\angle DQE$ is the average of $\widehat{DE}$ and $\widehat{CB}$, we have $\widehat{CB} = 30^\circ$. Now that we have $\widehat{CB}$, we can find $\angle CPD$:

$$\angle CPD = (\widehat{CD} - \widehat{CB})/2 = 20^\circ.$$

Since $\overline{DP}$ bisects $\angle CPE$, we know $\angle DPE = \angle CPD = 20^\circ$. Because $\angle DPE = (\widehat{DE} - \widehat{AB})/2$, we find $\widehat{AB} = 10^\circ$.

Now that we have all the other arcs of the circle, we can find our desired arc by subtracting from 360° : $360^\circ - 50^\circ - 70^\circ - 30^\circ - 10^\circ = 200^\circ$. $\square$

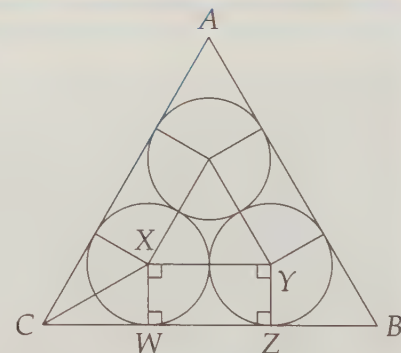
Concept: Label all the given information in the problem, and all the information you find as you find it. This will help you discover new facts about the problem. For example, your final diagram in the last problem might look like the figure below. Notice that our various arc measures and angles are marked, and that the equal angles at P are marked equal.



Problem 12.21: Three congruent circles with radius 1 are drawn inside equilateral $\triangle ABC$ such that each circle is tangent to the other two and to two sides of the triangle. Find the length of a side of $\triangle ABC$.

Solution for Problem 12.21: We need to create simple figures to work with, so we start by connecting the centers of our circles and drawing radii to tangent points. (Note that connecting the centers of the circles is the same as drawing radii to where the circles are tangent.) Since $WXYZ$ is a rectangle (because $XW = YZ$, $\overline{XW} \parallel \overline{YZ}$, and $\angle XWZ = 90^\circ$), we have $WZ = XY = 2$. Hence, we need only find CW to finish, since BZ is the same as CW .

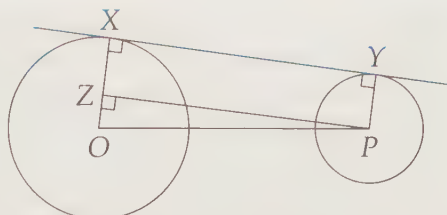
We draw $\overline{CX}$ to build a right triangle and note that this segment bisects $\angle ACB$ because circle X is tangent to both $\overline{AC}$ and $\overline{BC}$ (and hence its center is equidistant from them). Since $\angle XCW = (\angle ACB)/2 = 30^\circ$, $\triangle CXW$ is a 30-60-90 triangle. Thus, $CW = XW \sqrt{3} = \sqrt{3}$. Finally, we have $BC = BZ + ZW + WC = 2 + 2\sqrt{3}$. $\square$



Concept: When you have tangents in a problem, it's often very helpful to draw radii to points of tangency to build right triangles. When you have tangent circles, connect the centers. (In fact, if you have multiple circles in a problem, connecting the centers will sometimes help even when the circles aren't tangent.)

Problem 12.22: X is on circle O and Y on circle P such that $\overleftrightarrow{XY}$ is tangent to both circles. Given that $OP = 40$, and the radii of circles O and P are 31 and 7, respectively, find XY .

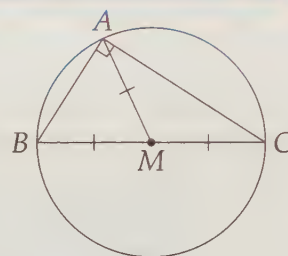
Solution for Problem 12.22: We start with the usual segments to draw: $\overline{XY}$, the radii to points of tangency, and the segment connecting the centers. We still don't have a right triangle to work with, but we do know that radii $\overline{OX}$ and $\overline{PY}$ are both perpendicular to tangent $\overline{XY}$ as shown. Since both $\overline{OX}$ and $\overline{PY}$ are perpendicular to the same line, they are parallel. We make a right triangle and a rectangle by drawing a line through P parallel to $\overline{XY}$. $ZX = PY = 7$, so $OZ = OX - ZX = 24$. Since $OP = 40$, we have $PZ = 32$ from right triangle $\triangle OZP$. Since $XYPZ$ is a rectangle, we have $XY = ZP = 32$, so the length of the common tangent is 32.



$\overline{XY}$ is called a common *external* tangent of the two circles. As an Exercise, you'll find the length of the common *internal* tangent, too. $\square$

Problem 12.23: Median $\overline{AM}$ of $\triangle ABC$ has length 8. Given that $BC = 16$ and $AB = 9$, find the area of $\triangle ABC$.

Solution for Problem 12.23: When we draw the figure and label all our lengths, we see that $BM = AM = CM = 8$. Therefore, a circle centered at M with radius 8 goes through all three vertices of $\triangle ABC$. Since $\overline{BC}$ is a diameter of this circle, $\angle BAC$ is inscribed in a semicircle and therefore must be a right angle. So, $AC = \sqrt{BC^2 - AB^2} = 5\sqrt{7}$. $\triangle ABC$ is a right triangle, so its area is half the product of its legs:



$$[ABC] = \frac{(AB)(AC)}{2} = \frac{(9)(5\sqrt{7})}{2} = \frac{45\sqrt{7}}{2}.$$

$\square$

Using this same reasoning, we can also prove this important fact:

Important:



If the length of a median of a triangle is half the length of the side to which it is drawn, the triangle must be a right triangle. Moreover, the side to which this median is drawn is the hypotenuse of the right triangle.

We can also look to this problem for some important problem solving techniques:

Concept:



When stuck on a problem, always ask yourself 'Where have I seen something like this before?' In Problem 12.23, we have a median that is half the side to which it is drawn. This should make us think of right triangles, since the median to the hypotenuse of a right triangle is half the hypotenuse. Then we go looking for right triangles.

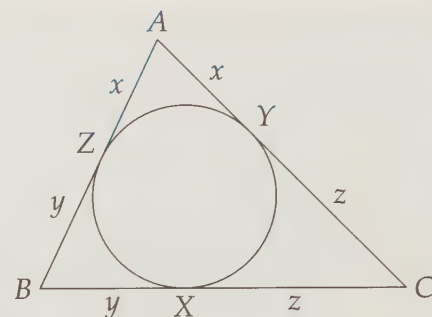
Concept: Always be on the lookout for right triangles.



Problem 12.24: Let Z be the point where the incircle of $\triangle ABC$ meets $\overline{AB}$. Find AZ in terms of the sides of $\triangle ABC$.

Solution for Problem 12.24: We start by labeling the equal tangents from the vertices as shown in the diagram. We want to relate x to the sides of the triangle, so we write the sides of the triangle in terms of x , y , and z . We let the sides of the triangle be $AB = c$, $AC = b$, and $BC = a$ and we have:

$$\begin{aligned} AB &= c = x + y \\ AC &= b = x + z \\ BC &= a = y + z \end{aligned}$$



We want x in terms of a , b , and c . Adding the three equations will give us $x + y + z$, which we can use with $y + z = a$ to find x :

$$a + b + c = 2(x + y + z) \quad \text{so} \quad x + y + z = \frac{a + b + c}{2}.$$

We can then subtract the equation $y + z = a$ from $x + y + z = (a + b + c)/2$ to find

$$x = \frac{a + b + c}{2} - a = s - a,$$

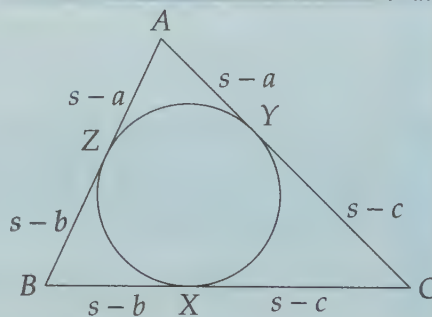
where s is the semiperimeter (half the perimeter) of the triangle. Similarly, $y = s - b$ and $z = s - c$. $\square$

Concept: Symmetric systems of equations can often be easily solved by either multiplying all the equations or adding them.

Important: The lengths from the vertices of $\triangle ABC$ to the points of tangency of its incircle are given as follows:

$$\begin{aligned} AZ &= AY = s - a \\ BZ &= BX = s - b \\ CX &= CY = s - c \end{aligned}$$

where $AB = c$, $AC = b$, and $BC = a$, and the semiperimeter of $\triangle ABC$ is s .



Exercises

Problems 12.4.3, 12.4.4, 12.4.5, and 12.4.8 are very important relationships that you'll be seeing again in your study of more advanced geometry. Be sure to pay special attention to them.

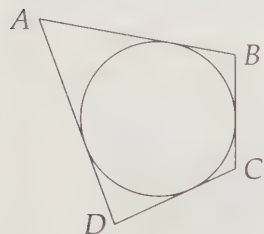
12.4.1 Is every quadrilateral cyclic?

12.4.2 Prove the following about cyclic quadrilaterals:

- A cyclic parallelogram must be a rectangle.
- A cyclic rhombus must be a square.
- A cyclic trapezoid must be isosceles.

12.4.3 Prove that if median $\overline{XM}$ of $\triangle XYZ$ has half the length of $\overline{YZ}$, then $\triangle XYZ$ is a right triangle with hypotenuse $\overline{YZ}$.

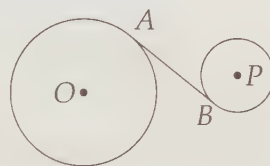
12.4.4 $ABCD$ is a cyclic quadrilateral. Prove that $\angle ACB = \angle ADB$.



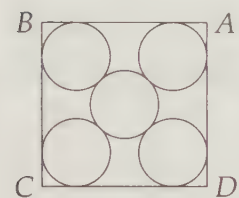
12.4.5 Quadrilateral $ABCD$ in the diagram at left is a **circumscribed quadrilateral**, meaning that it is circumscribed about a circle, so the circle is tangent to all four sides of $ABCD$. Show that $AB + CD = BC + AD$. **Hints:** 570

12.4.6 Does every quadrilateral have an inscribed circle (a circle tangent to all four sides), as $ABCD$ does in the previous problem?

12.4.7 In the figure at right, $\overline{AB}$ is tangent to both $\odot O$ and $\odot P$. The radius of $\odot O$ is 8, the radius of $\odot P$ is 4, and $OP = 36$. find AB . (A common tangent like $\overline{AB}$ is sometimes called the **common internal tangent** of two circles.) **Hints:** 591



12.4.8 Prove that the inradius of a right triangle with legs of length a and b and hypotenuse c is $(a + b - c)/2$.



12.4.9★ The five circles in the diagram are congruent and $ABCD$ is a square with side length 4. The four outer circles are each tangent to the middle circle and to the square on two sides as shown. Find the radius of each of the circles. **Hints:** 543, 416

12.5 Construction: Tangents

Now you'll use your newfound tangent knowledge to construct tangents to circles.

Problems

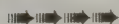
Problem 12.25: Given a circle with center O and point A on the circle, construct a line through A that is tangent to the circle.

Problem 12.26: Given a circle with center O and point P outside the circle, construct a line through P that is tangent to the circle.

Problem 12.27: Nonintersecting circles with centers O and P are shown. Construct a line that is tangent to *both* circles.



Extra! Seek simplicity, and distrust it.



—Alfred North Whitehead

We'll start with constructing a tangent through a point on a circle.

Problem 12.25: Given a circle with center O and point A on the circle, construct a line through A that is tangent to the circle.

Solution for Problem 12.25: Make sure you see why this is not a correct solution:

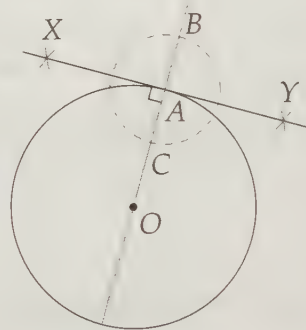
Bogus Solution: We place our straightedge on the paper so that it goes through the point A , then we slowly turn it, keeping it through A , until it only touches the circle in one point.



This Bogus Solution is *not* a Euclidean construction – it's just a sketch. We need to be able to prove our construction works. Here, the proof would essentially be 'Our line touches the circle once because we say so.' That's not good enough. We need to use geometric principles to prove our line is tangent, not just eyeball judgement.

We know how to construct perpendiculars, and we know that a tangent is perpendicular to the radius drawn to the point of tangency. Putting these together, we have a pretty straightforward construction. We draw $\overrightarrow{OA}$, which includes the radius to point A , then construct the line through A perpendicular to $\overline{OA}$. This line is our tangent line.

Our construction of this tangent line is indicated in the diagram at right. We draw a circle centered at A , which hits $\overrightarrow{OA}$ at B and C . We then draw two pairs of intersecting arcs with the same radius centered at B and C to find X and Y . $\overleftrightarrow{XY}$ is the perpendicular bisector of $\overline{BC}$, and is therefore our tangent line. $\square$

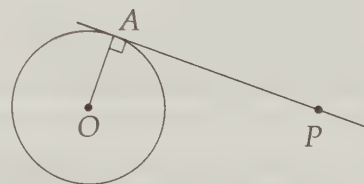


So we can handle a tangent through a point on a circle; how about one through a point outside the circle?

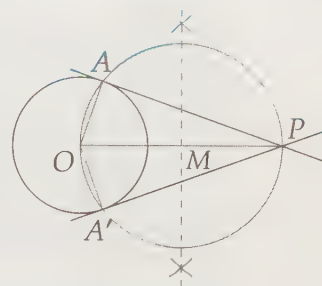
Problem 12.26: Given a circle with center O and point P outside the circle, construct a line through P that is tangent to the circle.

Solution for Problem 12.26: Unfortunately, we can't simply construct a radius, then draw a perpendicular line, because we don't know which radius to draw! So, feeling a little stuck, we look for a simpler problem to solve. But that doesn't get us anywhere, either – there's not any obvious simpler problem that will help us with this one. Therefore, we draw the completed diagram and look for clues how we might possibly construct it given only the circle and point P .

We go ahead and include the radius to the point of tangency because right angles are very useful, and we know how to construct perpendicular lines. If we can construct point A , where the tangent meets the circle, we'll be set. The only seemingly useful information we have is that $\angle OAP$ is a right angle, i.e., $\triangle OAP$ is a right triangle. So, we consider what we know about right triangles.



The circumcenter of any right triangle is the midpoint of its hypotenuse. Since A must be on the circumcircle of $\triangle OAP$, and $\triangle OAP$ must be a right triangle with hypotenuse $\overline{OP}$, we know that A is on the circle that has $\overline{OP}$ as a diameter. Therefore, to find A , we construct M , the midpoint of $\overline{OP}$. We then draw a circle with center M and radius MO . Where this circle hits $\odot O$ is the point A where our tangent touches the circle. $\overrightarrow{PA}$ is the tangent. (And the other point where the new circle hits $\odot O$ gives us the other tangent to $\odot O$ through P .) $\square$



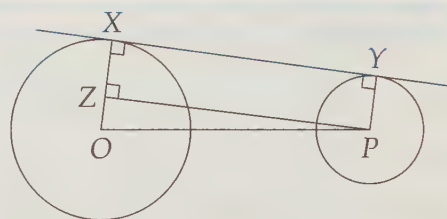
Concept: When tackling challenging construction problems, try drawing a completed diagram (without straightedge & compass construction). Then use observations about your completed diagram to figure out how to construct it with straightedge & compass alone.



Now that we've mastered tangent constructions, and have an intriguing new construction-finding strategy, let's try an even more challenging problem.

Problem 12.27: Given two nonintersecting circles with different radii, construct a line that is tangent to both.

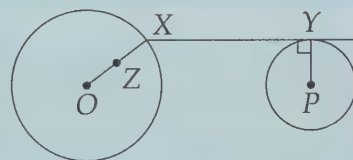
Solution for Problem 12.27: This construction looks pretty complicated, so we start from the completed diagram. We should probably add radii to the tangent points to give us perpendicular lines to work with. As we draw these, we realize, 'Hey, we've done a problem very much like this!' We think back to Problem 12.22, which we solved by creating the diagram shown at right.



Maybe this diagram will give us a clue. If we can find a way to construct any of X , Y , or even Z given only circles $\odot O$ and $\odot P$, we'll be able to construct a line tangent to both circles.

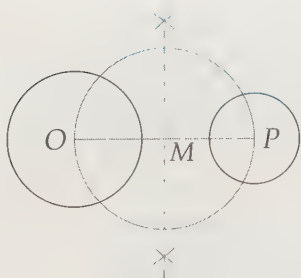
At this point, we might think:

Bogus Solution: Pick a point Z , draw $\overrightarrow{OZ}$ until it hits $\odot O$. That gives us point X . Then construct a line through X that is tangent to $\odot P$ using the construction from Problem 12.26.



Unfortunately, this badly fails. If we just take any old Z , we might get a diagram like the one we want, but we probably won't. $\overrightarrow{XY}$ is indeed tangent to $\odot P$, since we constructed it to be tangent to $\odot P$. However, this line is not necessarily tangent to $\odot O$, as our diagram clearly points out.

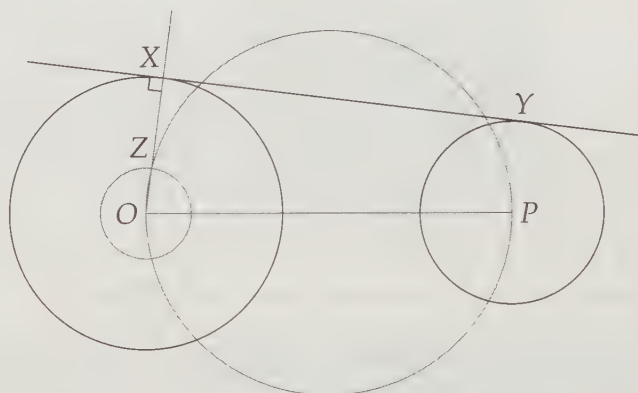
So, we can't just pick any point to be Z (or X , or Y). And we can't just keep trying different possibilities for Z until we get one that works! We must find a specific one that we can prove works. Looking again at our 'working backwards' diagram we drew first, we see that Z is the vertex of the right angle of right triangle $\triangle OZP$. As we saw in Problem 12.26, this means that Z is on the circle that has $\overline{OP}$ as a diameter.



So, starting from just our circles and their centers, we at least have a first step: we construct a circle with $\overline{OP}$ as a diameter. Point Z must be on this circle. If we can find another circle or line with Z on it, we'll be finished. Thinking one more time about Problem 12.26 and our 'working backwards' diagram, we note that OZ is the difference between the radii of the two circles. Call this difference d . We can construct a segment with length d since we know the radii of the circles!

After constructing a segment with length d , we can draw a circle with center O and d as the radius. Since Z must be d away from O , it must be on this little circle. Therefore, the intersection of this circle and the circle with $\overline{OP}$ as diameter is Z ! Extending $\overrightarrow{OZ}$ to hit $\odot O$ gives us X , and drawing the line perpendicular to $\overrightarrow{OZ}$ through X gives us the common tangent.

Of course, you'll get to prove that this final line is tangent to $\odot P$ as an Exercise. $\square$



Exercises

12.5.1 Given two intersecting lines, construct a circle that is tangent to both of them.

12.5.2 Prove that our construction in Problem 12.27 does produce a line that is tangent to both circles.

12.5.3★ In Problem 12.27, we constructed the common external tangent of two circles. How can we construct a common internal tangent of two nonintersecting circles (i.e., a line tangent to both circles that intersects the line connecting the centers of the circles)? **Hints:** 29, 440, 378

12.6 Summary

Definition: An angle formed by two chords of a circle is **inscribed** in the angle it cuts off.

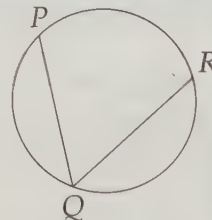
Important:



- Any angle inscribed in a semicircle is a right angle.
- An inscribed angle equals $1/2$ the measure of the arc it intercepts. For example,

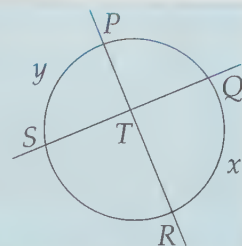
$$\angle PQR = \frac{\widehat{PR}}{2}.$$

- Any two angles that are inscribed in the same arc are equal.



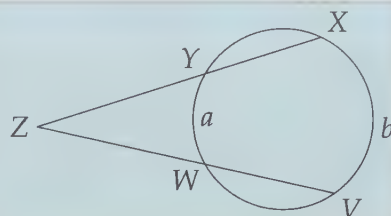
Important: The measure of the angle formed by two intersecting chords is the average of the measures of the arcs intersected by the chords. For example, in the diagram at right, we have

$$\angle PTS = \angle QTR = \frac{\widehat{PS} + \widehat{QR}}{2} = \frac{x + y}{2}.$$



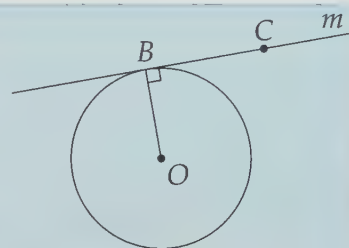
Important: Two secants that meet at a point outside a circle form an angle equal to half the difference of the arcs they intercept. For example, in the diagram we have

$$\angle Z = \frac{b - a}{2}.$$



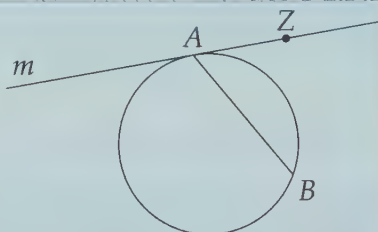
Important: A line tangent to a circle is perpendicular to the radius drawn to the point of tangency. Conversely, a line drawn through a point on a circle that is perpendicular to the radius drawn to that point must be tangent to the circle. For example, for the diagram at right, we can write:

$\overleftrightarrow{BC}$ is perpendicular to radius $\overline{OB}$ if and only if $\overleftrightarrow{BC}$ is tangent to circle O .



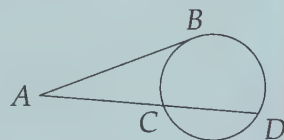
Important: An angle formed by a tangent and a chord with the point of tangency as an endpoint equals one-half the arc intercepted by the angle. For example, in the figure at right, line m is tangent to the circle at A , so

$$\angle ZAB = \frac{\widehat{AB}}{2}.$$



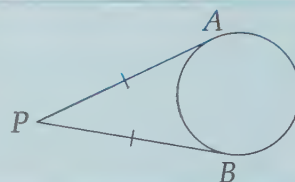
The angle formed by a tangent and a secant is half the difference of the intercepted arcs. For example,

$$\angle BAD = \frac{\widehat{BD} - \widehat{BC}}{2}.$$



Important:


As shown in the diagram at right, we can draw two tangent segments to a circle from a point outside the circle. These tangents are always equal to each other in length.


Important:


A **cyclic quadrilateral** is a quadrilateral that can be inscribed in a circle. The opposite angles of any cyclic quadrilateral sum to 180° .

Important:


If the length of a median of a triangle is half the length of the side to which it is drawn, the triangle must be a right triangle. Moreover, the side to which this median is drawn is the hypotenuse of the right triangle.

Important:

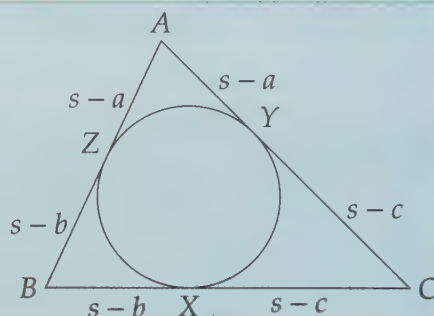

The lengths from the vertices of $\triangle ABC$ to the points of tangency of its incircle are given as follows:

$$AZ = AY = s - a$$

$$BZ = BX = s - b$$

$$CX = CY = s - c$$

where $AB = c$, $AC = b$, and $BC = a$, and the semiperimeter of $\triangle ABC$ is s .


Important:


The length of the inradius of a right triangle with hypotenuse of length c and legs of lengths a and b is

$$\frac{a + b - c}{2}.$$

Problem Solving Strategies

Concepts:


- When trying to prove something or find some general rule, investigating a few sample cases can be an excellent guide.
- When you can't find any more lengths or angles in a problem but you still haven't solved it, try assigning a variable to one of the lengths or angles. Then, find other lengths or angles in terms of that variable. Finally, label your diagram with everything you find – this will make it easier to see when you have enough information to solve the problem.

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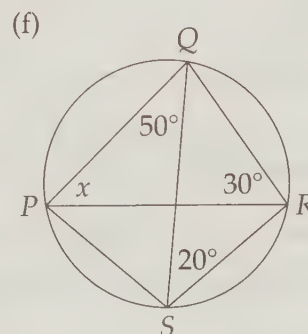
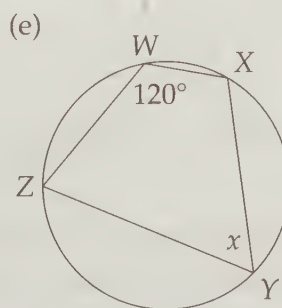
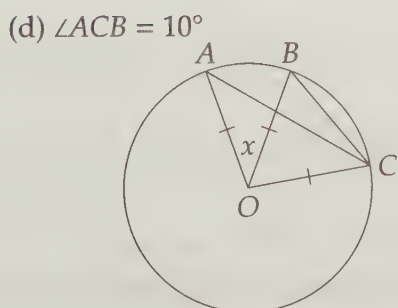
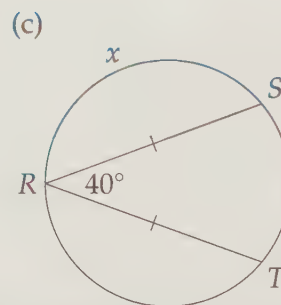
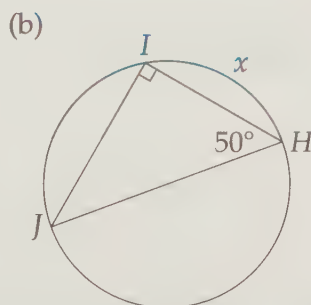
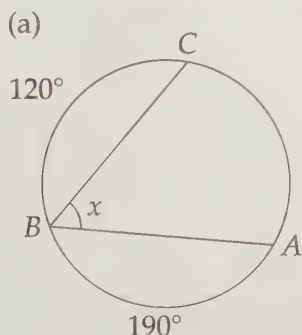
Concepts: ... continued from the previous page



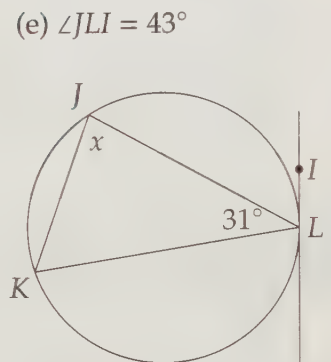
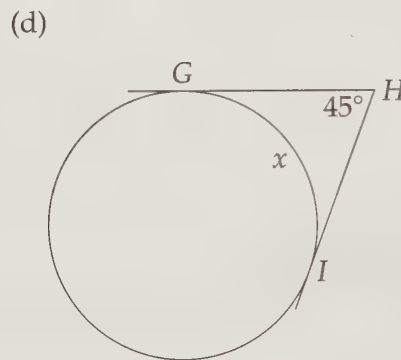
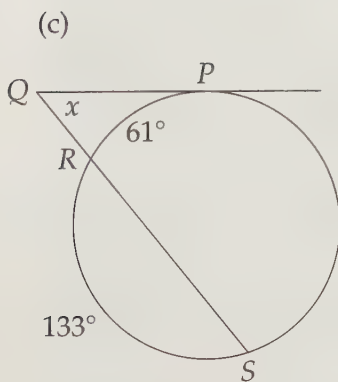
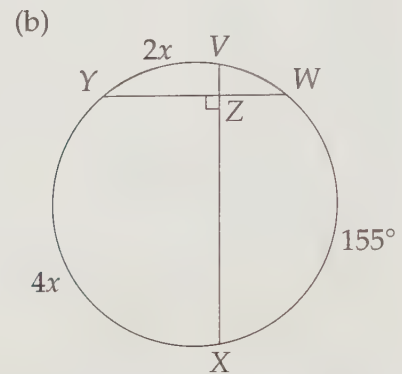
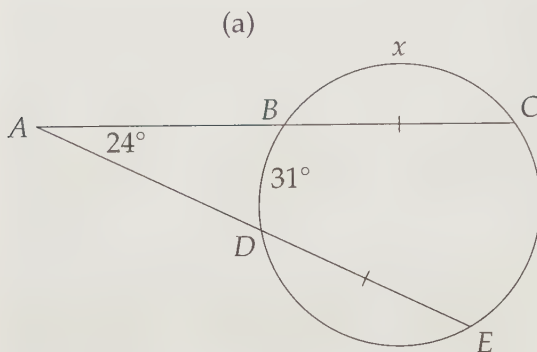
- When you have tangents in a problem, it's often very helpful to draw radii to points of tangency to build right triangles. When you have tangent circles, connect the centers. (In fact, if you have multiple circles in a problem, connecting the centers will sometimes help even when the circles aren't tangent.)
- When stuck on a problem, always ask yourself 'Where have I seen something like this before?'
- Always be on the lookout for right triangles.
- Symmetric systems of equations can often be easily solved by either multiplying all the equations or adding them.
- When tackling challenging construction problems, try drawing a completed diagram (without straightedge & compass construction), then using observations about your completed diagram to figure out how to construct it with straightedge & compass alone.

REVIEW PROBLEMS

12.28 Find x in each of the following

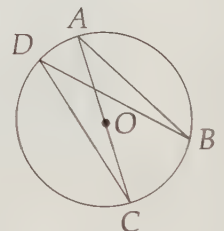


12.29 Find x in each of the following. (Assume that all lines that look like tangent lines are tangent lines.)



12.30 $\overline{AC}$ is a diameter of $\odot O$ in the diagram at left below. Given $\angle ACB = 58^\circ$ and $BD = CD$, find the following:

- $\widehat{AB}$.
- $\angle BAC$.
- $\angle DBC$.



12.31 Points A, B, C , and D are on a circle in that order, and $\overline{AC}$ and $\overline{BD}$ meet at E . Given that $\widehat{AB} = 34^\circ$, $\widehat{CD} = 102^\circ$, $\widehat{DA} = 109^\circ$, find $\angle BEC$.

12.32 Prove that if line m passes through a point B on circle O such that m is perpendicular to $\overline{OB}$, then m must be tangent to the circle. (Try this one without looking back in the book for the proof!) **Hints:** 302

12.33 Use your knowledge about inscribed angles to explain why the circumcenter of an obtuse triangle is outside the triangle.

12.34 Point T is on minor arc $\widehat{RS}$ of $\odot O$. Given that $\angle ROS = 53^\circ$, find $\angle RTS$.

12.35 $\overline{KT}$ meets $\odot T$ at Y . Given that $YK = 6$ and $YT = 2$, find the length of the tangent segment from K to $\odot T$.

12.36 Point Y is on $\odot F$ and X outside $\odot F$ such that $\overline{XY}$ is tangent to F . $\overleftrightarrow{XF}$ intersects $\odot F$ at A and B such that $XA < XB$. Given that $\widehat{YB} = 134^\circ$, find $\widehat{AY}$, $\angle YXF$, and $\angle XYA$.

12.37 $\overline{AT}$ and $\overline{BT}$ are tangent to $\odot O$ as shown at left below. Given $AT = 6$ and $\widehat{AB} = 120^\circ$, find OT , $\angle ATB$, and $[OATB]$.

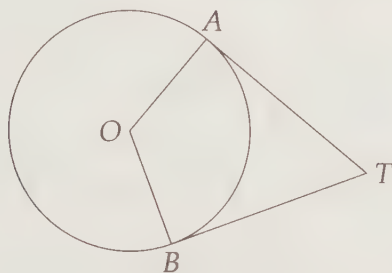


Figure 12.7: Diagram for Problem 12.37

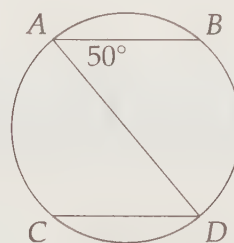


Figure 12.8: Diagram for Problem 12.38

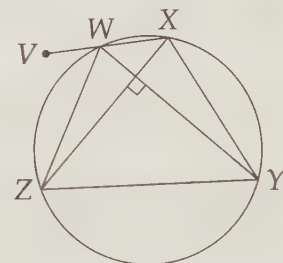
12.38 In the circle at right above, $\overline{AB} \parallel \overline{CD}$, $\overline{AD}$ is a diameter of the circle, and $AD = 36$. What is the length of $\overline{AB}$? (Source: MATHCOUNTS)

12.39 $ABCDEFGHIJKL$ is a regular dodecagon. Use the circumcircle of the dodecagon to find the following:

- $\angle ABC$.
- $\angle ACD$.
- $\angle ADJ$.
- the acute angle between $\overline{AF}$ and $\overline{BI}$.

12.40 Given $\angle VWZ = 81^\circ$ and $\angle WZX = 38^\circ$ in the diagram at right, find the following:

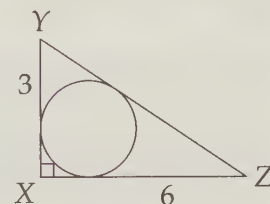
- $\angle VXZ$.
- $\angle WYZ$.
- $\angle XZY$.
- $\angle XYZ$.
- $\angle WXY$.



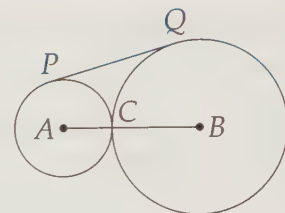
12.41 $ABCD$ is a square with area 100. $\odot O$ is tangent to all four sides of the square. Diagonal $\overline{BD}$ meets the circle at X and Y , with X closer to D than to B . Find DX .

12.42 Right triangle $\triangle XYZ$ and its incircle are shown at right. As shown, Z is 6 units from the point where the incircle touches $\overline{XZ}$, and Y is 3 units from the point where the incircle touches $\overline{XY}$. Find YZ .

12.43 Prove that the incircle of a given triangle is unique. (In other words, prove that there is only one circle tangent to all three sides of a given triangle.)



12.44 Circles $\odot A$ and $\odot B$ are tangent at point C . P is on $\odot A$ and Q is on $\odot B$ such that $\overline{PQ}$ is tangent to both circles. Given $AC = 3$ and $BC = 8$, find PQ .



12.45 A circle is inscribed in a regular hexagon and another circle is circumscribed about the hexagon (i.e. it passes through all six vertices of the hexagon). Find the ratio of the area of the smaller circle to the area of the larger circle.

Hints: 195

Challenge Problems

12.46 In the diagram at left below, lines ℓ and m are tangent to the circle. Prove that $\ell \parallel m$.

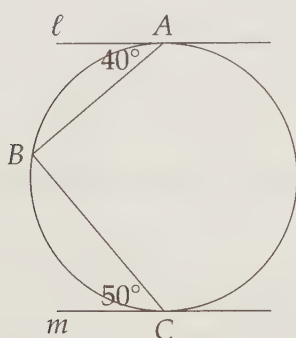


Figure 12.9: Diagram for Problem 12.46

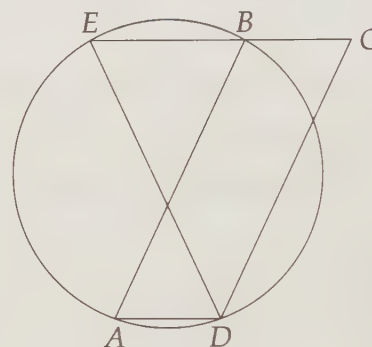


Figure 12.10: Diagram for Problem 12.47

12.47 $ABCD$ is a parallelogram. $\overline{BC}$ extended past B meets the circumcircle of $\triangle ABD$ at point E as shown at right above. Prove that $ED = CD$.

12.48 $\triangle ABC$ is a right triangle and the radius of its inscribed circle equals 5. If the perimeter of $\triangle ABC$ exceeds twice its hypotenuse by N , compute N . (Source: ARML) **Hints:** 449

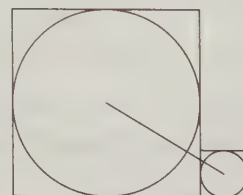
12.49 Let A, B, C , and D be points on a circle ω , in that order. Let P and Q be points on ω such that $\overline{PA}$ bisects $\angle DAB$ and $\overline{QC}$ bisects $\angle BCD$. Prove that $\overline{PQ}$ is a diameter of ω .

12.50 Let the incircle of triangle ABC be tangent to sides $\overline{BC}$, $\overline{AC}$, and $\overline{AB}$ at points D , E , and F , respectively. Given that $\angle A = 32^\circ$, find $\angle EDF$. **Hints:** 127

12.51 $\overline{AB}$ is a diameter of $\odot O$ and $\overline{AX}$ and $\overline{AY}$ are chords of the circle such that $AX = AY$. Prove that $\overline{AB}$ is the perpendicular bisector of $\overline{XY}$. **Hints:** 210, 289

12.52 Find XZ in the right triangle described in Problem 12.42. **Hints:** 352

12.53 The areas of two adjacent squares are 256 square inches and 16 square inches, respectively, and their bases lie on the same line. What is the number of inches in the length of the segment that joins the centers of the two inscribed circles? (Source: MATHCOUNTS) **Hints:** 55, 456



12.54 Given a circle of radius 2, there are infinitely many line segments of length

2 that are tangent to the circle at their midpoints. Find the area of the region consisting of all such line segments. (Source: AMC 12) **Hints:** 49

12.55 Let C_1 and C_2 be two circles that intersect at two points, A and B . Let P be the point diametrically opposite A on C_1 , and let Q be the point diametrically opposite A on C_2 . Prove that P , B , and Q are collinear. **Hints:** 87

12.56 Chord $\overline{CE}$ is a segment of the perpendicular bisector of radius $\overline{OA}$ of $\odot O$. $\overline{AX}$ is a diameter of $\odot O$. Find $\angle OCX$. **Hints:** 264

12.57 Three of the sides of $ABCD$ are tangent to the circle shown at left below. Given that $AB = 6$, $BC = 7$, $CD = 8$, and $AD = 9$, find $CY + DX$. (Source: Mandelbrot) **Hints:** 173

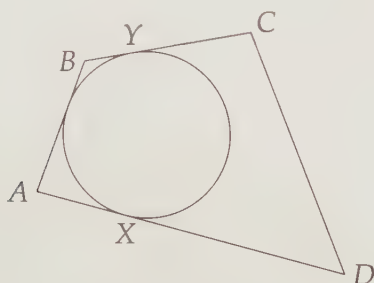


Figure 12.11: Diagram for Problem 12.57



Figure 12.12: Diagram for Problem 12.58

12.58 A room is built in the shape of the region between two semicircles with the same center and parallel diameters such that the smaller semicircle is contained within the larger. The room is shown at right above. The farthest distance between two points with a clear line of sight is 12m. What is the area in m^2 of the room? (Source: HMMT) **Hints:** 316

12.59 The incircle of $\triangle ABC$ is tangent to $\overline{AB}$ at Z . Show that if the inradius of $\triangle ABC$ equals AZ , then the angle $\angle BAC$ is a right angle. **Hints:** 348

12.60 The sides of right triangle $\triangle ABC$ all have integer lengths. Prove that the inradius of $\triangle ABC$ also has integer length. **Hints:** 449

12.61 Let ω be the circumcircle of triangle ABC . Let A_1 , B_1 , and C_1 be the midpoints of arcs $\widehat{BC}$, $\widehat{CA}$, and $\widehat{AB}$, respectively, of circle ω . In this problem we prove that the incenter of triangle ABC is the orthocenter of triangle $A_1B_1C_1$.

- Let I be the incenter of $\triangle ABC$. Why must $\overrightarrow{AI}$ pass through A_1 ?
- Why must $\overline{AA_1}$ be perpendicular to $\overline{B_1C_1}$? **Hints:** 256
- Combine the previous parts to conclude that the incenter of triangle ABC is the orthocenter of triangle $A_1B_1C_1$.

12.62★ Prove that quadrilateral $ABCD$ is cyclic if $\angle A + \angle C = 180^\circ$. (Note that this is the converse of what we discussed in Problem 12.19.) **Hints:** 331, 37

12.63★ We are given points A , B , C , and D in the plane such that $AD = 13$ while $AB = BC = AC = CD = 10$. Find $\angle ADB$. (Source: Mandelbrot) **Hints:** 82, 199

12.64★ Let $\overline{PQ}$ be a diameter of a circle and T be a point on the circle besides P and Q . The tangent to the circle through point Q intersects $\overrightarrow{PT}$ at R , and the tangent through T intersects $\overline{QR}$ at M . Prove that M is the midpoint of $\overline{QR}$. **Hints:** 222, 128

Extra! Most construction problems involve the two classical tools of geometry, the straightedge and compass. These tools can solve an array of construction problems, many of which are discussed in this book, but perhaps even more interesting are construction problems that are impossible to solve with these tools.

Three famous classical construction problems that the ancient Greeks couldn't solve are listed below.

(a) Squaring the Circle: Given a circle, construct a square of equal area.

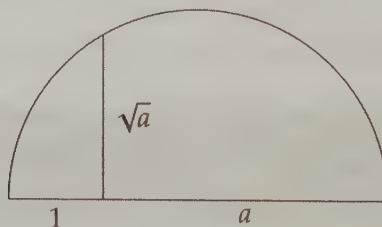
(b) Doubling the Cube: Given a cube, construct another cube with twice the volume of the first cube.

(According to Greek legend, the Athenians appealed to the Oracle for advice about a plague that was ravaging their city. The Oracle's answer was to have their cube-shaped altar to the god Apollo doubled in size. The Athenians then attempted to follow this advice by building a new altar that had twice the side length of the original altar, but this had the effect of increasing the volume of the altar by a factor of $2^3 = 8$, not 2, and as a result, the plague worsened.)

(c) Trisecting the Angle: Given an arbitrary angle, divide it into three equal angles.

We now know that the ancient Greeks couldn't solve these problems because all three of these constructions are impossible to perform with straightedge and compass.

Modern mathematicians proved these constructions are impossible by determining what lengths could be constructed if we are given segments of lengths a , b , and 1. With these segments as a starting point, the lengths $a + b$, $a - b$, ab , and a/b can be constructed. See if you can figure out how to construct ab and a/b . Even $\sqrt{a}$ can be constructed, as the diagram below suggests. (See if you can figure out how this construction works!)



This means that any of the following lengths can be constructed:

$$\sqrt{2}, \quad \frac{1 + \sqrt{5}}{2}, \quad \sqrt{\frac{\sqrt{3} + 8\sqrt{7}}{2}} - \sqrt{5} + \sqrt[4]{17 - \sqrt{3 + \sqrt{2}}}.$$

Continued on the next page. . .

Extra! . . . continued from the previous page



In other words, any length involving the integers and the arithmetic functions $+$, $-$, $\times$, $\div$, and $\sqrt{}$ can be constructed. Furthermore, these are the only lengths that can be so constructed. So how can this fact be used to solve the three problems?

The first problem, squaring the circle, is equivalent to constructing $\sqrt{\pi}$. However, it was shown by Lindemann in 1882 that π is transcendental, meaning that π is not the root of any polynomial with integer coefficients, which also implies the same for $\sqrt{\pi}$. Since every constructible number is the root of some polynomial with integer coefficients, we know that neither π nor $\sqrt{\pi}$ can be constructed.

The second problem, doubling the cube, is equivalent to constructing $\sqrt[3]{2}$. A branch of mathematics, known as Galois theory, tells us that $\sqrt[3]{2}$ cannot be expressed using arithmetic functions and square roots of integers. Hence, $\sqrt[3]{2}$ cannot be constructed.

For the third problem, trisecting the angle, certain angles can be trisected. For example, given a straight angle of 180° , a third of it would be 60° , which can be easily constructed. But can a 60° angle itself be trisected? In other words, can a 20° angle be constructed? It can be shown that if $x = \cos 20^\circ$, then $8x^3 - 6x - 1 = 0$. Again, Galois theory tells us that such an x cannot be expressed using arithmetic functions and square roots. This counterexample of trisecting the 60° degree angle shows that trisection of angles cannot be done in general.

Despite the fact that these problems have been conclusively solved for years, many amateur mathematicians still attempt to find constructions for these problems and submit them to mathematical journals, hoping for fame. All these attempts are completely futile.

The same Galois theory that rules out these constructions also tells us which regular polygons can be constructed. The construction of an equilateral triangle and a regular hexagon are easy, but beyond that, which other regular n -gons are constructible?

To answer this question, we introduce the **Fermat numbers**. The k^{th} Fermat number is $F_k = 2^{2^k} + 1$, so $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, and so on. A **Fermat prime** is a Fermat number that is also prime. At the time of writing, there are only five known Fermat primes, namely F_0, F_1, F_2, F_3 , and F_4 .

A regular n -gon is constructible if and only if n is a power of 2 or is of the form

$$n = 2^j F_{m_1} F_{m_2} \cdots F_{m_i},$$

where j is an arbitrary non-negative integer and $F_{m_1}, F_{m_2}, \dots, F_{m_i}$ are *distinct* Fermat primes. So the regular polygon with $3 \cdot 17 = 51$ sides is constructible, but the regular polygons with 7 sides and 9 sides are not. Gauss first proved that the regular 17-gon is constructible, and was said to have been so proud of this fact that he requested that a regular 17-gon be etched into his tombstone.

It is surprising that either one of the straightedge or compass by itself can essentially do the same work as both of them put together. The **Mohr-Mascheroni Theorem** states that any construction that can be performed by straightedge and compass can be performed by compass alone. If the target of a construction is a line, the compass alone cannot produce the line; however, it can produce two points on the line, which defines the line correctly. Analogously, the **Poncellet-Steiner Theorem** states that any construction that can be performed by straightedge and compass can be performed by straightedge alone, as long as a single circle and its center are given.